



# Experimental and kinetic modeling study of butene isomer pyrolysis: Part I. 1- and 2-Butene



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## ABSTRACT

1- and 2-Butene pyrolysis experiments were performed using a tubular flow reactor at an absolute pressure of  $\sim 0.82$  atm over a temperature range of 535–810 °C with residence times ranging from  $\sim 0.5$ – $\sim 2.4$  s. The initial concentration of the fuel ranged from 5 to 50 mol%. These data were compared to the predictions of a fundamentally based detailed kinetic model. For both sets of experiments, the model accurately predicts the observed fuel conversion, production of light products, and the formation of several important molecular weight growth species. The primary pathways that lead to the fuel decay and major product formation are discussed. One of the important kinetic features of 1- and 2-butene pyrolysis is that significant amounts of 1,3-butadiene are produced. The addition reactions of vinyl, allyl, and methyl allyl radicals to 1,3-butadiene are all energetically favorable, leading to substantial production of higher molecular weight species. The model was also applied to published pyrolysis and oxidation data for 1- and 2-butene that were collected under significantly different conditions. The comparison to the latter data sets illustrate that, even under oxidation conditions, it is important to properly account for the pyrolysis kinetics.

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## 1. Introduction

Significant progress has been achieved in characterizing the kinetics of hydrocarbon fuels. These efforts have identified approaches to optimize energy conversion processes that use these fuels. Alkanes comprise a substantial fraction of hydrocarbon fuels, and their kinetics are quite well understood. The initiation process usually involves C–C bond fission (except methane), producing radicals that subsequently undergo  $\beta$ -scission reactions to generate smaller radicals and alkenes. The smaller radicals will then abstract from the fuel, accelerating its consumption. These reactions and their associated rate coefficients are rather well characterized. Kinetic models are available that describe the pyrolysis, oxidation, and/or combustion of alkanes. Representative examples include the Livermore [1], the JetSurF [2], and the CRECK [3] mechanisms. More information can be found in two review articles [4,5]. In contrast, the subsequent kinetics of the alkenes is less well-known. Alkenes are also components in hydrocarbon fuels, and they are also key intermediates during the decomposition of alkanes, alcohols, ethers, and biofuels, and therefore their

subsequent reactions will have substantial impact on the final product distribution. More specifically, alkenes, unlike alkanes, can undergo radical addition reactions that then may lead to formation of higher molecular weight growth (MWG) species, perhaps leading to soot formation. Therefore, any accurate kinetic model capable of describing alkanes must contain reliable alkene kinetics to properly capture the kinetics at higher levels of conversion, especially the reactions involved in MWG formation.

Butene is the smallest alkene with isomeric structures. With four heavy atoms (i.e., non-hydrogen atoms), the butene isomers serve as reasonable model compounds for larger olefins, while being small enough to be amenable to analysis by high-level electronic structure calculations. The study of 1-butene ( $\text{C}=\text{CCC}$ ,  $1\text{-C}_4\text{H}_8$ ) and 2-butene ( $\text{CC}=\text{CC}$ ,  $2\text{-C}_4\text{H}_8$ ) pyrolysis kinetics can be used to characterize the impact of the location of the double bond. One interesting complication that arises when considering kinetics of these butene isomers is the potential impact of molecular isomerization reactions. Consider the initiation reaction for 1-butene and 2-butene. 1-Butene dissociation to form allyl + methyl is expected to be faster than 2-butene dissociation to form methyl-allyl + H, given the relative strengths of the allylic C–C versus allylic C–H bonds. However, if the isomerization reaction of 2-butene to 1-butene were fast, then the two dissociation rates may be more similar.

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The pyrolysis, oxidation, and combustion of 1- and 2-butene [6–21] have been the focus of many studies. Several studies have shown that the pyrolysis kinetics is faster under certain conditions than the oxidation kinetics. The implication of this is that the products that are formed by the pyrolysis of the parent fuel dictate the initial conditions for the oxidation kinetics. For instance, in a recent study by Fenard et al. [14,15] of the oxidation of 1- and 2-butene, the experimental observations at lower temperatures clearly showed that there was substantial fuel conversion prior to the consumption of oxygen. Thus, an accurate description of the pyrolysis chemistry is essential for the proper description of oxidation and/or combustion kinetics. A similar conclusion was reached by Pan et al. [16] who measured the ignition delay times of 1-butene behind reflected shock waves; they used three published models, NUIG Aramco Mech 1.3 [22], USC Mech 2.0 [2], and LLNL C<sub>4</sub> model [1], to predict ignition delay times. They found that none of these models could reproduce the experimental data, especially at lower temperatures. Similar observations have been made in alkane oxidation studies [23,24]. These findings provide the motivation to develop an improved butene pyrolysis mechanism.

One of the most important kinetic features for alkene pyrolysis at ~1000 K is that resonantly-stabilized free radicals (RSFRs) are likely to accumulate to very high concentrations [25,26]. Due to the lower energies for allylic C–C and/or C–H bond dissociation, RSFRs are the likely initial dissociation products as well as being more likely to be formed by hydrogen abstraction reactions. Due to their stability, the concentrations of RSFRs are generally higher than those of “normal” (i.e., non-resonantly stabilized) alkyl radicals. A variety of reactions involving RSFRs that lead to the formation of MWG species during alkene pyrolysis have been investigated. These include H-shift, cyclization, addition, recombination, and H-abstraction reactions [25,27–33]. In particular, the initially-formed adducts that result from olefin addition or from radical recombination after a subsequent H-abstraction, can isomerize to form cyclic radicals that may turn into stable cyclic species after the  $\beta$ -scission of H-atoms or small alkyl groups. These stable cyclic products are important precursors for the formation of 1,3-cyclopentadiene and benzene, the building blocks for polyaromatic hydrocarbons (PAH) generation. Some H-abstraction reactions of alkenes by resonantly-stabilized radicals are approximately thermoneutral, since they form the conjugate alkene and a different stabilized radical. This type reaction may lead to a redistribution of RSFRs, thus impacting the various pathways leading to MWG species.

The objective of this work is to develop a fundamentally-based kinetic model that can describe the pyrolysis of the 1- and 2-butene, including the fuel conversion and formation of light and MWG products. A companion paper examines the pyrolysis kinetics of isobutene [34]. The model is a slight variation of the one that was used to characterize propylene pyrolysis [26]. Comprehensive flow reactor pyrolysis experiments were performed at an absolute pressure of ~0.82 atm; the temperature (535–810 °C), extent of inert gas dilution (50, 10, and 5 mol % in N<sub>2</sub>) and residence time (~0.5, 1.2, and 2.4 s) were varied. The fuel conversions ranged from virtually no reaction to over 96% and 89% for 1- and 2-butene, respectively. The light products and multiple MWG products were quantified. The measured mole fractions exhibit consistently good carbon and hydrogen mass balances. The kinetic model was able to accurately predict the fuel conversions and formation of light species as well as many MWG products. The model was also applied to published pyrolysis data [12] that were collected under significantly different conditions and to relatively low temperature oxidation data [14,15] to illustrate the importance of accurately characterizing the pyrolysis kinetics.

## 2. Methods

### 2.1. Experimental specifications

Experiments were conducted using a continuous flow tubular reactor. Details of the apparatus and procedure have previously been discussed [26], and therefore only a brief description is given here. A known flow rate of the fuel, either 1-butene (Matheson TriGas Co., 99.9%) or 2-butene (Matheson TriGas Co., 99.9%), was mixed with N<sub>2</sub> (General Air, 99.998%) and introduced into a 6 mm ID heated tubular quartz reactor that was housed in an electric furnace. The axial temperature profile of the reactor was measured using a K-type thermocouple. The reproducibility of the measurements are  $\pm 2$  °C in the center, constant temperature region of the reactor. At the edges of the reactor, where the temperature is lower, the error in the measurement is higher. All the temperature profiles are provided in Table S1 in the Supporting information (SI). The temperature in the center region of the reactor is approximately constant, and falls off steeply at the edges of the reactor. Experiments were conducted at ambient (high-altitude) pressure of ~0.82 atm. The initial concentration of 1-butene (5, 10, or 50 mol% in N<sub>2</sub>) and 2-butene (5 or 50 mol % in N<sub>2</sub>) was varied. A trace amount of isobutane and/or n-butane was measured in the 1- and 2-butene gas cylinders as impurities, and this was accounted for in the modeling efforts. The total flow rate was set to 30, 60, or 150 SCCM, which corresponds to nominal residence times of approximately 2.4, 1.0, and 0.5 s, respectively, for the center constant temperature region of the reactor (i.e., the hot zone). A summary of the experiment conditions for each dataset are provided in Tables S2 and S3 for 1- and 2-butene pyrolysis, respectively.

The reaction gas stream was analyzed using gas-chromatography (GC). Permanent gases and light hydrocarbons were detected using a thermal conductivity detector (TCD). Light and heavy hydrocarbons were detected using a flame ionization detector (FID). For the operating condition with initially 50 mol% fuel input at ~2.4 s, two different sets of 1- and 2-butene pyrolysis experiments were considered for model validation. In addition to that reported in this work, we include the earlier measurements by Al-Shoaibi and Dean [11]. A major improvement in the current experiments was that the column effluent was also measured by a mass spectrometer that was connected in parallel to the FID; this allowed for a more definitive product identification. The response factors were calibrated using commercial and in-house-prepared standard gas and liquid mixtures. The response factors of species that were not calibrated directly were estimated based on a correlation of response factors to the number of carbon atoms in the product for FID measurements. One limitation to the analytical method is that 1-butene and isobutene, 1,3-methylcyclopentadiene and 2,4-methylcyclopentadiene, as well as 1,3-cyclohexadiene and 1,4-cyclohexadiene are not completely separated from one another and, in these cases, the sum of the two isomers is reported. Since the response factors for these sets of species are similar, the impact on the total mass balance is negligible. For the case of 1-butene and isobutene, the relative amount of isobutene is small compared to the amount 1-butene. This will be discussed in more detail below.

### 2.2. Kinetic modeling

The detailed kinetic mechanism that was used in this study is very similar to the mechanism that was recently used to describe propylene pyrolysis [26]. The detailed development procedures for this manually-constructed mechanism are provided therein, thus only a brief description is provided here. It is a fundamentally-developed kinetic model. The rate constants are determined using

transition state theory based on the results of electronic structure calculations. In many cases, this approach is used to develop rate estimation rules that can be applied to larger species that are not calculated directly. However, when reliable experimental rate constant data are available, these values are implemented in the mechanism. The most significant modification to that mechanism is that the reaction pathways and corresponding rate constants for the  $C_4H_8$  potential energy surface (PES) have been updated. Several other modifications that are focused on the treatment of rate expressions to account for the pressure-dependent reactions were made; these include the dissociation of vinyl radical, the dissociation of cyclopentadiene, and the reactions involving  $C_4H_6$  species. Another modification involved correcting the rate constants for some reactions that are included in the mechanism, but which had no impact on the current predictions. Most of the pressure-dependent rate coefficients are written in Chebyshev form [35] and are applicable of a wide range of temperature and pressure (which are specified for each reaction). However, some of the less important pressure dependent reactions (e.g., the reactions on the  $C_3H_4$  PES) are described in terms of several different rate coefficients, each for a specified pressure. For these reactions, the user needs to specify the appropriate pressure. All these changes are described in the comments at the beginning of the mechanism (and the thermodynamic file) that are provided as text files in the SI. The mechanism does not distinguish between the cis- and trans-isomers of 2-butene, even though the isomers are separately measured, and therefore the predictions are compared to the sum of the two isomers.

Simulations incorporating the measured temperature profile were performed using the plug flow reactor module in ChemKin Pro [36]. This program was used to generate concentration-time/temperature profiles, rates of production plots, and sensitivity analysis. In many instances, the mechanism was run in an irreversible format (i.e., the forward and reverse reactions are both explicitly specified) to allow for partially equilibrated reactions to be identified. Note that in either format, reversible or irreversible, the reverse rate constants are calculated from the thermodynamic database and both formats provide identical concentration-time/temperature profiles.

In several places throughout the text, tables, and figures, we employ a chemical notation that omits the hydrogen atoms. For  $C_3$  and larger species, the radical site is indicated by “•”, a double bond by “=”, a triple bond by “≡”, and a cyclic species by “cy”.

### 3. Results

The results for all sets of experiments are provided in the SI (Tables S4–S10 for 1-butene and Tables S11–S15 for 2-butene). Each data point is an average of at least three measurements. The standard deviations of the measurements for concentrations above 0.1 mol% are typically ~0.5% of the average value; for the species at lower concentrations, the deviation usually ranges from ~2–6% of the average. It is important to note that the change in the measured mole fractions (with temperature or residence time) is due to both reaction and dilution since the total number of moles changes. The impact of dilution can be assessed from the change in the  $N_2$  mole fraction, since the number of moles of  $N_2$  remains constant. The majority of the products were identified. For the species that remain unidentified, the molecular formula is known from the mass spectrum. For each experiment, the measured mole fractions generally sum to one within the expected uncertainty of the experiments ( $\pm 0.02$ ). This suggests that we are quantitatively measuring all of the gas mixture that is contained in the GC sample loop. The measured carbon and hydrogen atom balances are also approximately unity ( $\pm 0.02$ ). This means that essentially all of the products that are produced in the reactor, even the heavy ones,

are making it to the GC sample loop. These results provide confidence that we are properly accounting for the vast majority of the reaction products.

The fuel decomposition for 1-butene ( $C=CCC$ ) and 2-butene ( $CC=CC$ ) pyrolysis is shown as a function of temperature at three different nominal “hot-zone” residence times in Fig. 1 for the data sets where the initial fuel concentration was 50 mol% in  $N_2$ . Both the cis- and trans-isomers of 2-butene were measured; however, only the sum is shown. These results include earlier measurements [11] at ~2.4 s using a simpler version of the flow reactor apparatus under similar conditions. The two sets of data are in good agreement, substantiating the reproducibility of the experimental system. The thermal decomposition of 1-butene starts at ~50 °C lower temperatures than that for 2-butene. The model predictions are also shown and they are in very good agreement for both butenes. As mentioned above, if any isobutene is formed during 1-butene pyrolysis, its peak will be hidden under the 1-butene peak. However, the model predicts 0.15% of isobutene at the highest 1-butene conversion, suggesting virtually no impact on the reported 1-butene mole fraction.

The formation of the light and MWG species per carbon atom is shown as a function of the fuel conversion for 1- and 2-butene pyrolysis at ~2.4 s for an initial fuel input of 50 mol% in Fig. 2. The earlier results that were collected using a simpler version of the flow reactor are also plotted along with the model predictions. Some interesting differences between the two fuels emerge. For the light  $\leq C_4$  products (left panels), methane is the dominant product at high conversion in both, but with slightly higher selectivity for 2-butene pyrolysis. The production of  $C_3$  products are similar for the two fuels. Significantly more  $C_2$  products are produced during 1-butene pyrolysis, while more  $C_4$  products are produced for 2-butene pyrolysis, and these products start forming at lower conversion levels. The model accurately captures the formation trends for the light species.

The initial MWG species (middle panels) are formed at lower levels of conversion for 1-butene pyrolysis. The  $C_5$  species' selectivity is higher for 1-butene pyrolysis and then declines at higher conversions, while those for 2-butene pyrolysis are still increasing. For both isomers, the  $C_6$  species selectivity are comparable and continue to increase with conversion. After a rapid early rise, the  $C_7$  selectivity in 1-butene pyrolysis remains approximately constant over a wide conversion range, while that for 2-butene pyrolysis increases with conversion. At the highest conversions the selectivities become comparable. The model predictions for the  $C_5$  to  $C_7$  species selectivities capture the impact of increasing conversion quite well, especially the differences observed for  $C_5$  and  $C_7$ .

The right hand panels show an increased selectivity of  $C_8$  species for 1-butene pyrolysis at low conversion. In general, there is a stepwise decrease in selectivity as the molecular weight of the products increases, and for all these species the selectivity increases rapidly with extent of conversion. Note the mole fractions of these species are an order of magnitude lower than those in the middle panels. The model consistently underpredicts the  $C_8$  selectivity for 1-butene pyrolysis, but it is reasonable for 2-butene pyrolysis. Conversely, the predictions for  $C_9$  and  $C_{10}$  are reasonable for 1-butene pyrolysis, while the  $C_9$  selectivity is underpredicted for 2-butene pyrolysis.

Figures 3 and 4 show the formation of light products as a function of temperature at three different residence times for the data sets where the initial fuel mole fraction was 50 mol% in  $N_2$  for 1- and 2-butene pyrolysis, respectively. Again, the plots also include the results from the earlier experiments [11] and, where comparisons can be made, the two data sets are in good agreement. The pyrolysis of both fuels produce the same light products. The major light products are methane, propylene, 1,3-butadiene, ethylene, and hydrogen. Additionally, 2-butene is produced from 1-butene

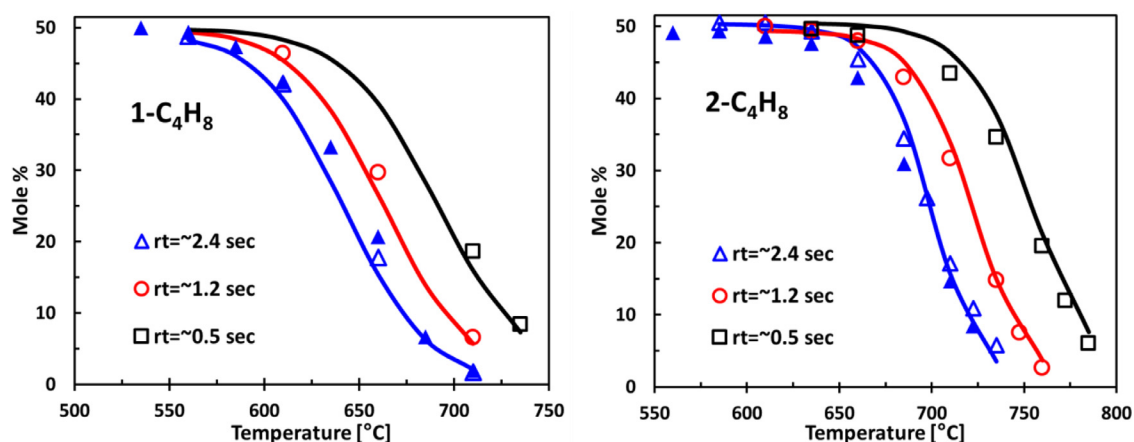


Fig. 1. Comparisons of measured fuel conversion to predictions of 1-butene and 2-butene versus temperature at three different residence times with initially ~50 mol% fuel (filled symbols—measured by Al Shoaibi and Dean [11]; open symbols—measured in this work; solid lines—predictions).

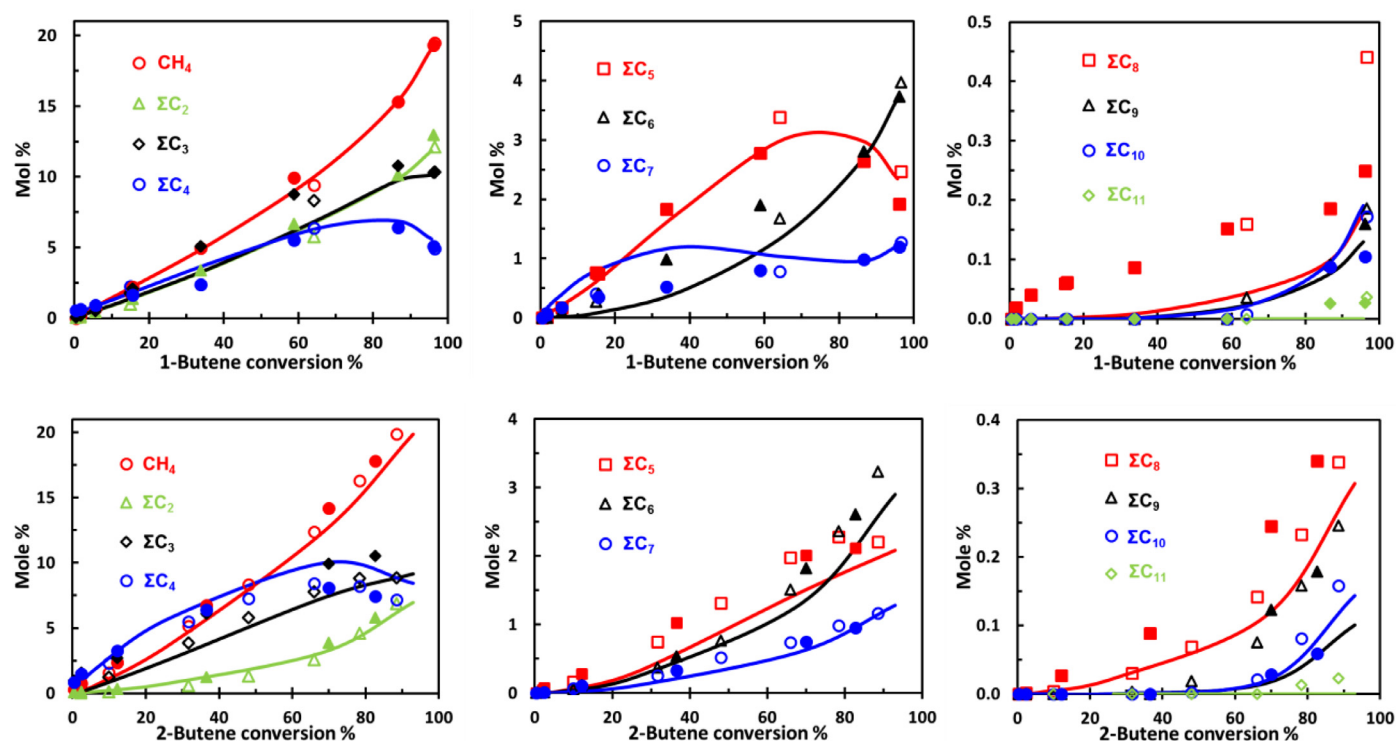


Fig. 2. Comparison of observed product distributions from the pyrolysis of 1-butene (top) and 2-butene (bottom) as a function of the fuel conversion. The initial fuel is ~50 mol% and the residence time is ~2.4 s; (filled symbols—Al Shoaibi and Dean [11]; open symbols—this work; lines—predictions).

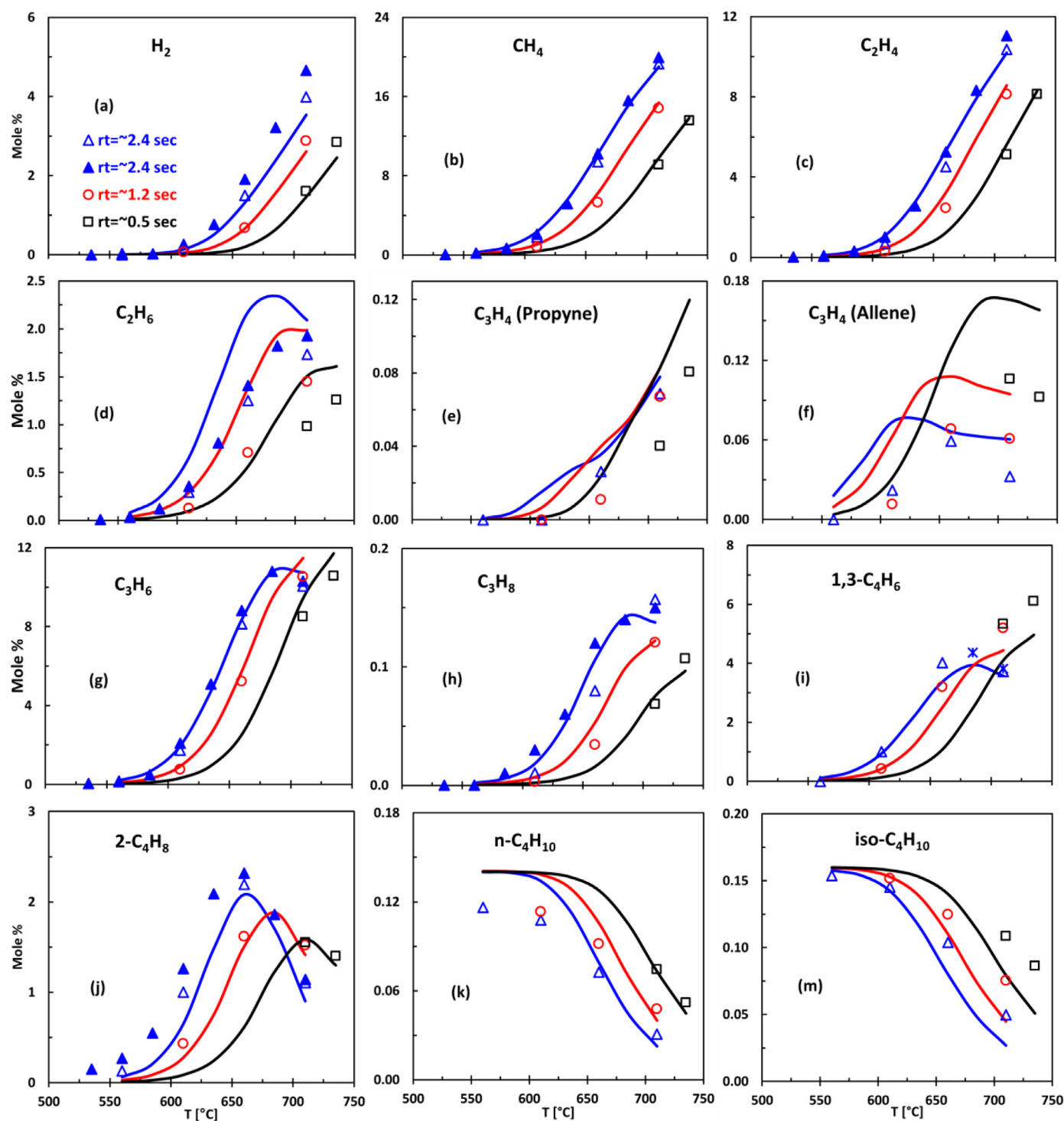
pyrolysis, while 1-butene is produced from 2-butene pyrolysis. Minor amounts of ethane, propyne, allene, and propane are produced. Isobutane and/or n-butane are contained as impurities in the fuel. For 2-butene pyrolysis, the data suggests that a very small amount of n-butane is produced before it decomposes.

The model predictions are also shown in Figs. 3 and 4. The model provides very good predictions for the effect of both temperature and residence time on formation of the major light products (hydrogen, methane, ethylene, propylene, and 1,3-butadiene) for both fuels. For 1-butene pyrolysis, the predictions for the formation of 2-butene are also quite good. For 2-butene pyrolysis, the formation of 1-butene is over predicted, although the model does predict the time and temperature dependence satisfactorily. Again, the model predicts that only a small amount of isobutene is formed in 2-butene pyrolysis (typically predicted to be less than 1% of the 1-butene concentration). The predictions for the minor light

products are also generally quite good. Allene is overpredicted for 1-butene pyrolysis, but is reasonable for 2-butene pyrolysis. It is encouraging to see that the model properly captures the markedly different effect of temperature and residence time on the allene mole fraction for the two isomers. The propyne predictions for 2-butene pyrolysis are low, but the trends with both temperature and residence time are captured. For 2-butene pyrolysis, the concentration of the n-butane impurity increases slightly before decreasing; the model fails to capture this trend, although the amount that was formed was extremely small. For 1-butene pyrolysis, the n-butane profile as well as that of the other impurity, isobutane, are accurately captured. Further comparisons of the model predictions for the fuel conversion and formation of the light species from all data sets are provided in parity plots in the SI (Figs. S1–S2).

Many MWG products were detected, and the most important ones are shown in order of decreasing production in Figs. 5





**Fig. 3.** Comparison of the observed and predicted mole percentages for the light products that are formed during 1-butene pyrolysis at three different residence times. The initial fuel input is 50 mol%. (filled symbols—Al Shoaibi and Dean [11]; open symbols—this work; solid lines—predictions)

and 6 for 1- and 2-butene pyrolysis, respectively. For the most part, the thermal decomposition of the two fuels leads to the same MWG products. The three major MWG products are benzene, toluene, and 1,3-cyclopentadiene. The measured mole fractions for these species increase as a function of temperature over the investigated temperature range. The other detected  $C_5$  products are: 2-methyl-1,3-butadiene ( $C=C_2C=C$ ), 2-pentene ( $CC=CCC$ ), 1,3-pentadiene ( $C=CC=CC$ ), 2-methyl-2-butene ( $C_2C=CC$ ), cyclopentenene ( $cyC_5H_8$ ), 1,4-pentadiene ( $C=CCC=C$ ), 2-methyl-1-butene

( $C=C_2CC$ ), and 1-pentene ( $C=CCCC$ ). The formation of several of these species agrees with the results of Schenk et al. [13] for the premixed flames for 1- and 2-butene and by Fenard et al. [14,15] for the oxidation of these two fuels. The other detected  $C_6$  products include the three methyl-cyclopentadiene isomers, two cyclohexadiene isomers, two methyl-cyclopentene isomers, cyclohexene, methyl-pentene isomers ( $C=CCC_2C$ ,  $C=CC_2CC$ , and  $CC=C_2C$ ), and several other unidentified  $C_6H_{10}$  species. The pyrolysis of 1-butene also forms 1,4-hexadiene ( $C=CCC=CC$ ). The other

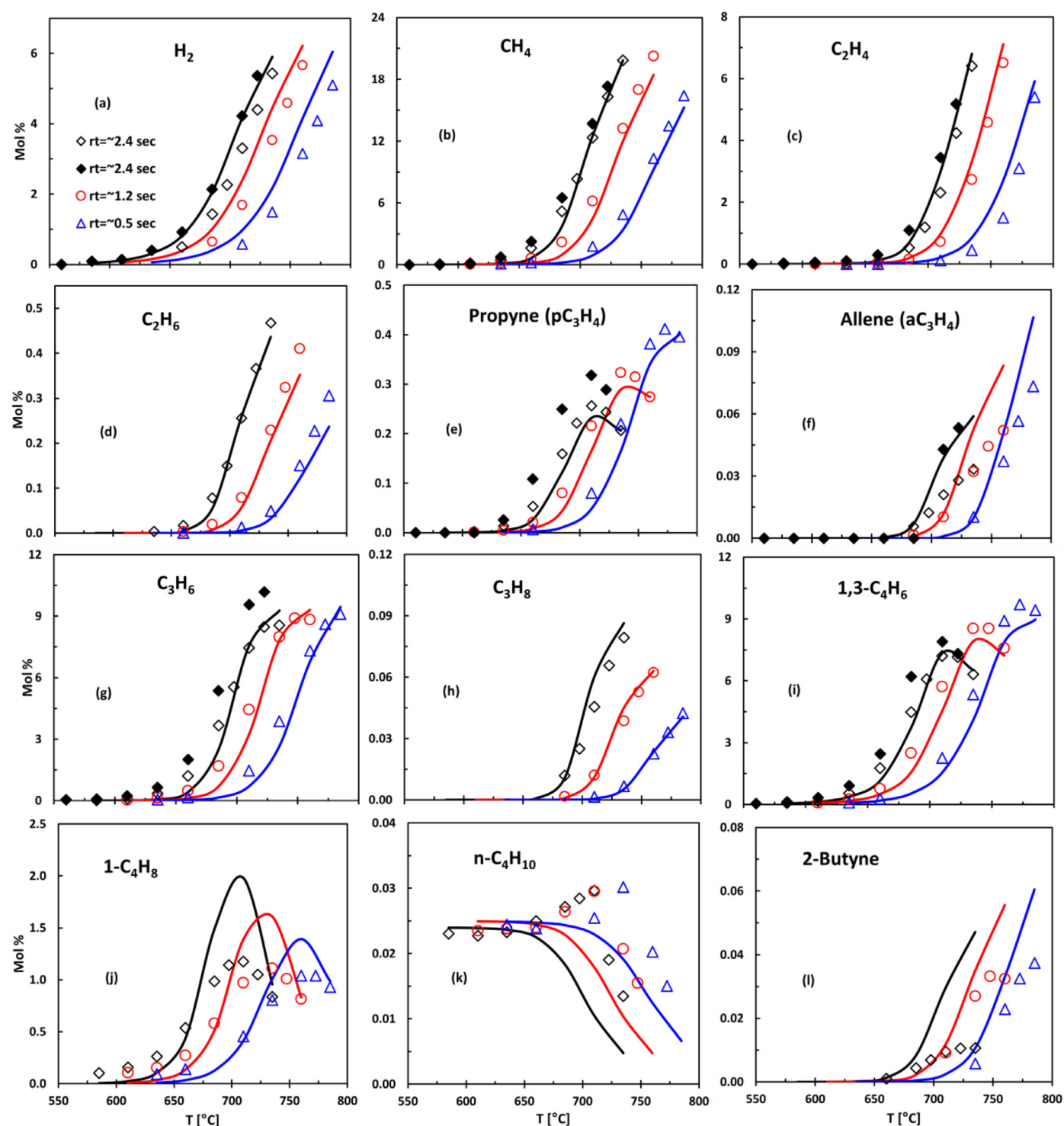


Fig. 4. Comparison of the observed and predicted mole percentages for the light products that are formed during 2-butene pyrolysis at three different residence times. The initial fuel input is 50 mol%. (filled symbols—Al Shoaibi and Dean [11]; open symbols—this work; solid lines—predictions)

detected  $C_7$  species are the methyl-cyclohexadiene isomers, ethyl-cyclopentene isomers, 1,3-cycloheptadiene, 1,3,5-cycloheptatriene, and several unidentified  $C_7H_{10}$  species. The concentration profiles for most of these  $C_5$ – $C_7$  species increases as the temperature increases, reach a maxima, and then begin to decrease at higher temperature. Styrene and ethylbenzene, indene, and naphthalene are the primary  $C_8$  to  $C_{10}$  species, respectively. The mole fractions of these species increase as a function of temperature over the investigated temperature range. Multiple minor products are also

identified, but are not shown in the figures. These include xylene isomers, ethenyl-cyclohexene, indane, propenyl-benzene, methyl-indene, 1,4-dihydro-naphthalene, and methyl-naphthalene. Some very small peaks are not unambiguously identified, due to their very low concentrations. For these cases, only the chemical formulas and retention times are provided in the SI.

As shown in Fig. 5, the model provides very good predictions for both temperature and time dependence for benzene and toluene production from 1-butene pyrolysis. A discrepancy for the

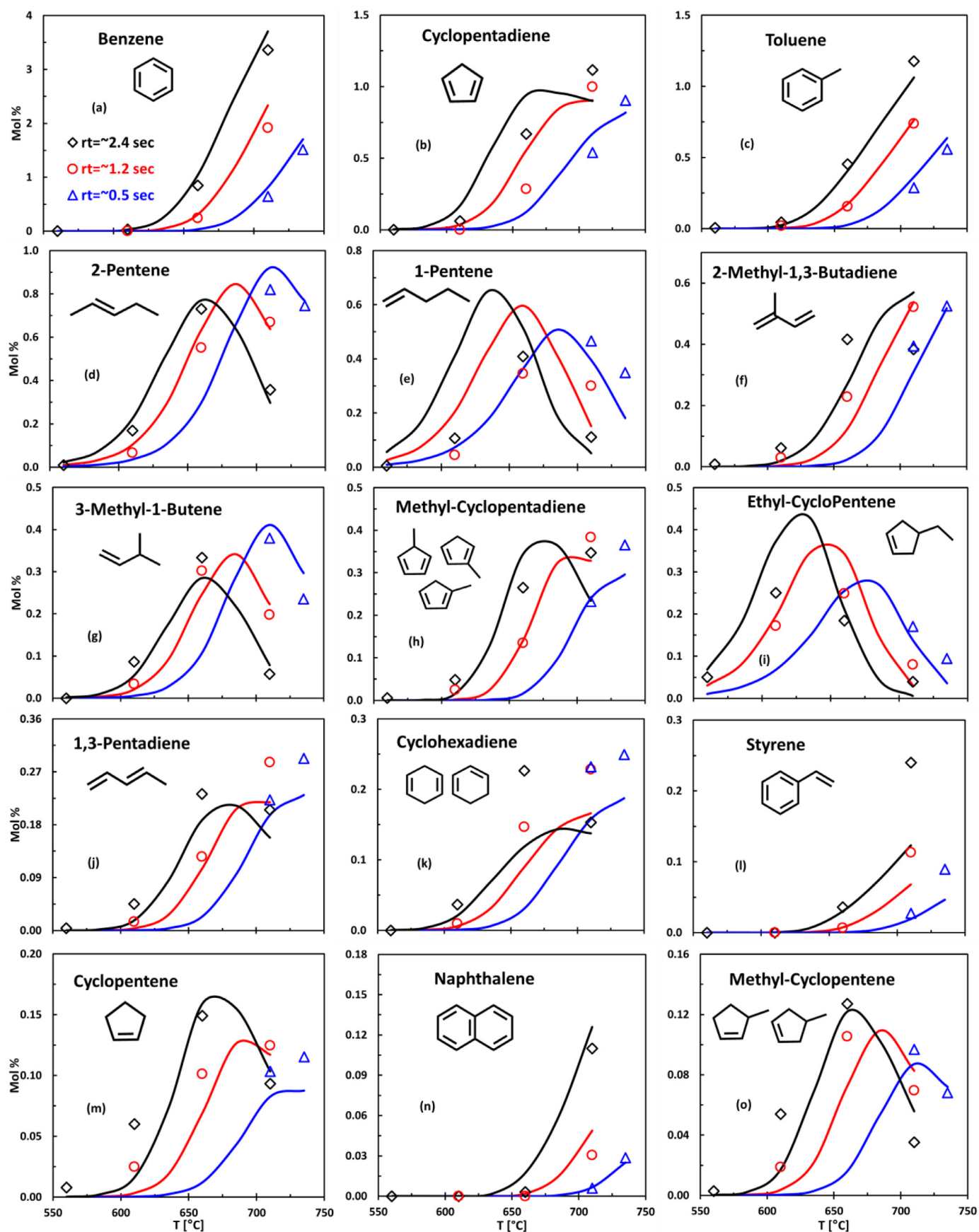


Fig. 5. Comparison of the observed and predicted mole percentages for the MWG products that are formed during 1-butene pyrolysis at three different residence times. The initial fuel input is 50 mol% (filled symbols—Al Shoaibi and Dean [11]; open symbols—this work; solid lines—predictions)

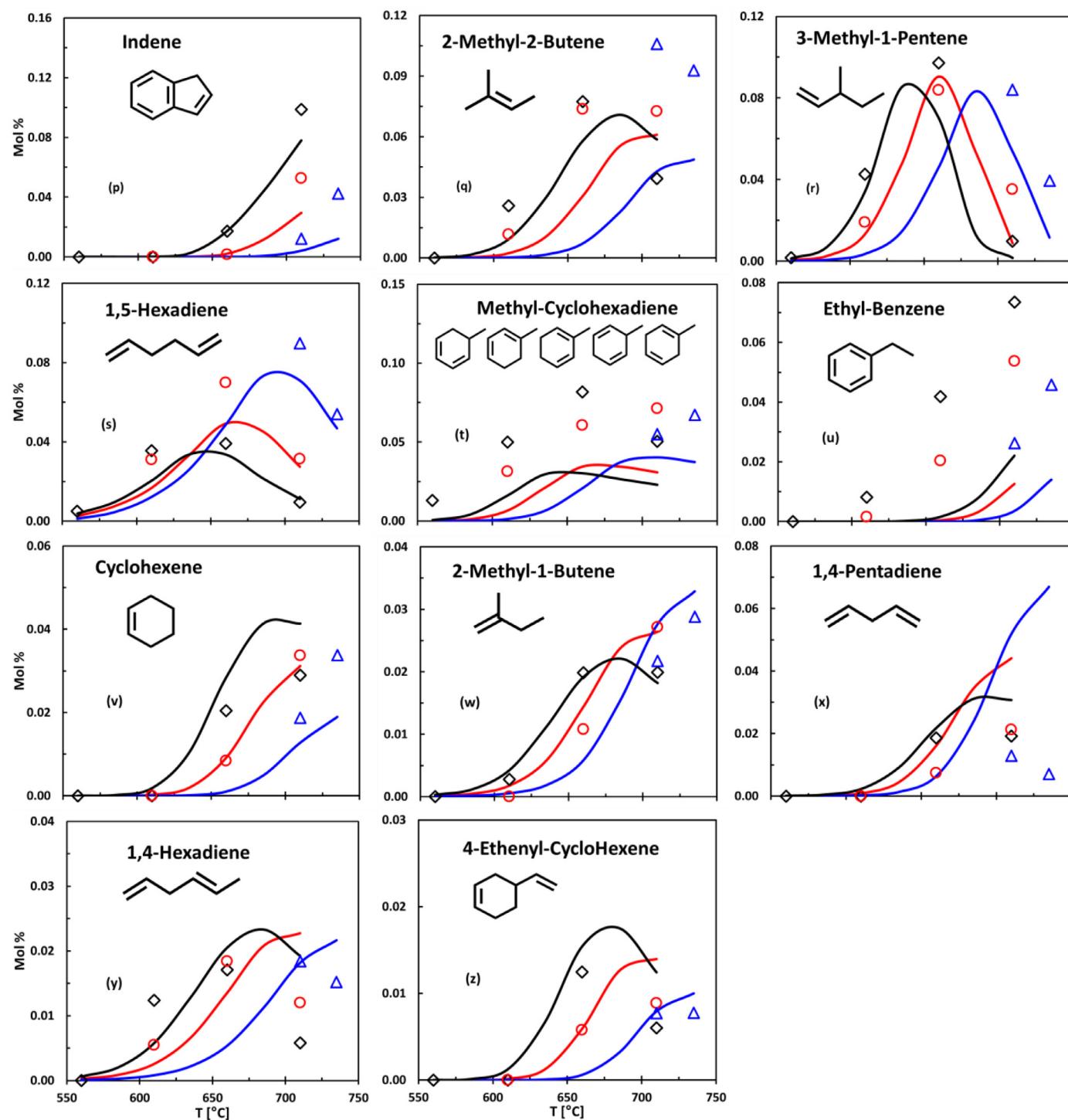


Fig. 5. Continued

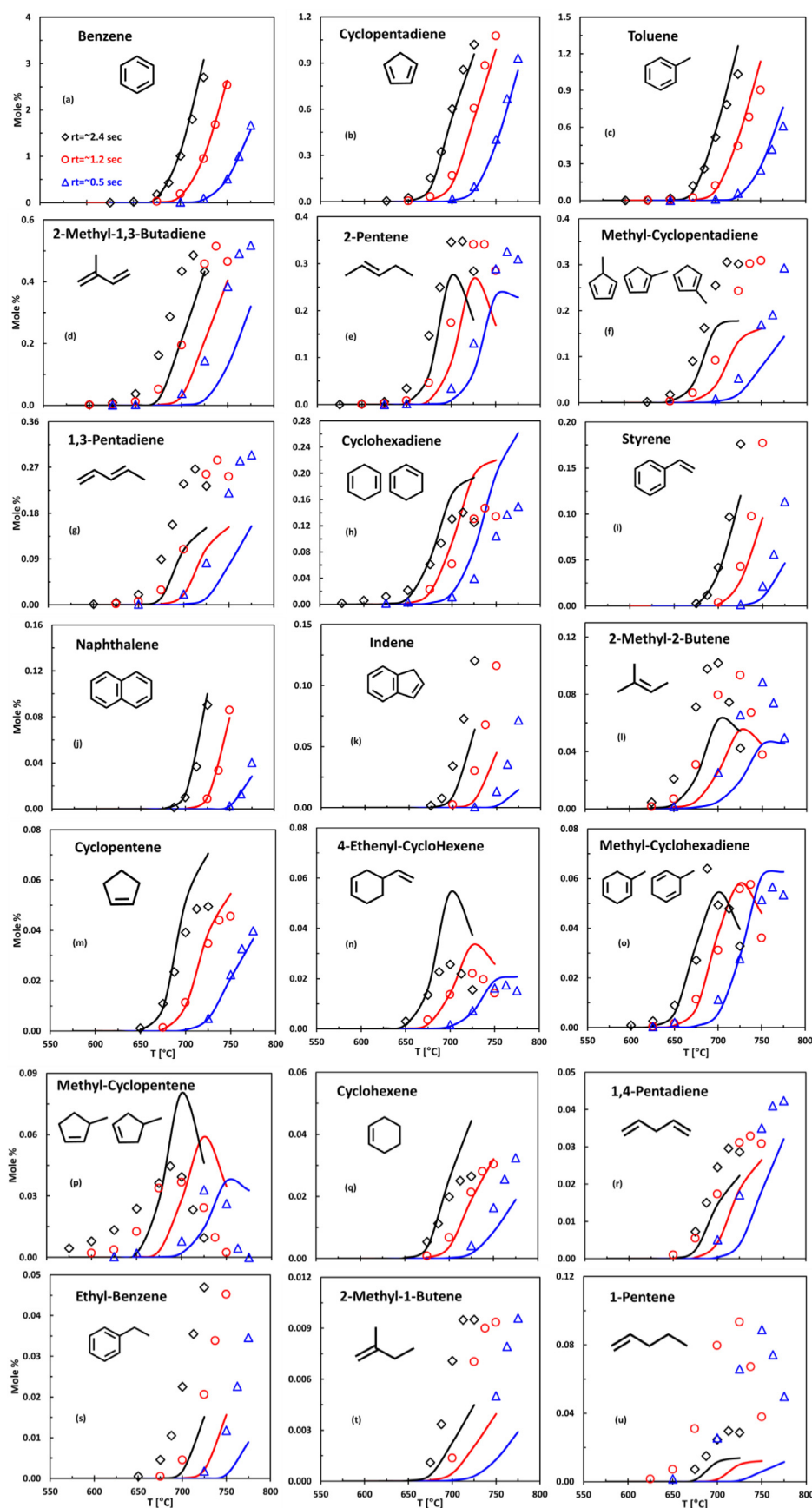
1,3-cyclopentadiene predictions is observed at the longer residence times, where the model production levels off at higher temperatures while the data do not. The model captures the trends for most of the minor species. The biggest discrepancy is ethylbenzene, where the model predictions are much lower than observed.

Figure 6 compares the predictions for 2-butene pyrolysis. The temperature and time dependence for formation of the major MWG species, benzene, 1,3-cyclopentadiene, and toluene, is very well described by the model. Although the predictions for the next four species in order of decreasing mole fraction (2-methyl-1,3-butadiene, 2-pentene, methylcyclopentadiene, and 1,3-pentadiene) are lower than observed by a factor of two or less, the trends with

time and temperature are properly described. Predictions for the next four species (cyclohexadiene, styrene, naphthalene, and indene) are quite good. The remaining species, at even lower mole fractions, are generally within a factor of two or so, with the exception of ethylbenzene which is lower by a factor of four to five. (Comparison of the model predictions to the MWG species data collected for all the experimental conditions in this study are shown as parity plots in the SI, Figs. S3 and S4.)

Overall the current mechanism, with no attempts to adjust the rate constants, provides a reasonable description of both reactant conversion, light products formation, and production of major MWG products for either 1- or 2-butene pyrolysis over a wide





**Fig. 6.** Comparison of the observed and predicted mole percentages for the MWG products that are formed during 2-butene pyrolysis at three different residence times. The initial fuel input is 50 mol% (filled symbols—Al Shoaibi and Dean [11]; open symbols—this work; solid lines—predictions)

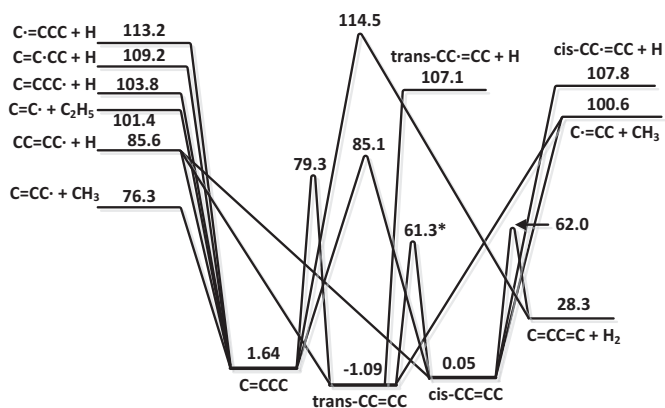


Fig. 7. The  $C_4H_8$  PES calculated at the CBS-QB3 level of theory. The barrier for isomerization of the trans- and cis-isomers of 2-butene, indicated by the asterisk, is taken from experiments (see text). Enthalpies are provided at 298 K in kcal/mol.

range of conditions at pressures near one atmosphere. We will show later that it also nicely describes literature data that were collected under significantly different conditions. An encouraging aspect of the comparisons is that many of the minor MWG product profiles are also captured by this model. Since this model was developed by assembling theoretically plausible reaction pathways and the associated rate coefficients, the next section will discuss the important reaction pathways during pyrolysis of the linear butene isomers.

#### 4. Discussion

In this section we discuss the key pathways that lead to the decomposition of 1- and 2-butene, the formation of the light products, and the formation of two major MWG products, benzene and 1,3-cyclopentadiene. The analysis presented here will focus on the data collected at the longest “hot-zone” residence time ( $\sim 2.4$  s) with the 710 °C and 735 °C temperature profiles (Table S1) the 1- and 2-butene pyrolysis, respectively. Under these conditions, the conversion of the two fuels is comparable ( $\sim 96\%$  for 1-butene and  $\sim 89\%$  for 2-butene) and substantial amounts of MWG species are formed. Most of discussion will focus on the kinetics that occurs in the constant temperature region of the reactor. However, it is important to keep in mind that initiation chemistry occurs near the front of the reactor where the temperature is much lower; at the end of the reactor the chemistry is quenched when the temperature decreases.

##### 4.1. Kinetics of fuel conversion

The thermal decomposition of 1- and 2-butene is initiated via reactions that occur on the  $C_4H_8$  potential energy surface (PES) shown in Fig. 7. The absolute energy for the stable species, radicals, and transition states were obtained mostly from CBS-QB3 calculations. An exception to this is the barrier for the cis- to trans-isomerization for 2-butene, which is based on extensive experiments [37–43]. The rate constants for the barrierless recombination reactions are estimated from the literature [44–46], and the rate constants for the reverse reactions are obtained based on the calculated equilibrium constants. The calculated rate constant for cis-2-butene forming 1,3-butadiene +  $H_2$  is consistent with the experiment measurements [18,37], as shown in Fig. S5. The calculated barrier for 1-butene forming 1,3-butadiene +  $H_2$  ( $E_a = 114.5$  kcal/mol) is consistent with that of ethane forming ethylene +  $H_2$  ( $E_a = \sim 114.0$  kcal/mol) [47].

On the  $C_4H_8$  surface, shown in Fig. 7, we have explicitly separated the cis- and trans-isomers of 2-butene. However, these are

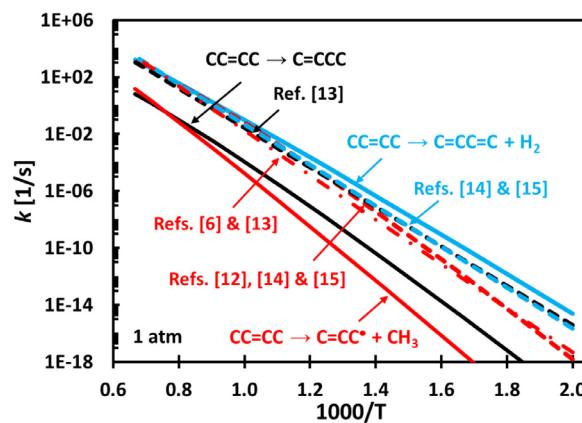


Fig. 8. Comparison of the current predictions (solid lines) for the unimolecular reactions of 2-butene to the literature reported values.

not distinguished in the kinetic mechanism. In the SI, Fig. S6, we show the numerical integration of the  $C_4H_8$  rate constants for the full versus a lumped cis- and trans-surface. The results show that the two isomers are quickly interconverted and the total 2-butene concentration is comparable for both surfaces. Thus, on the time scale of these experiments, the model predictions are not compromised by the use of the simplified (lumped)  $C_4H_8$  PES in the mechanism. The mechanism is also used to model experimental data that was collected at lower pressures, higher temperatures, and shorter time scales. Again, the model predictions are not significantly impacted by the use of simplified surface.

The rate constants for the unimolecular reactions of 1- and 2-butene that are used in the mechanism at 1 atm are compared to literature values in Fig. 8. The rate constants for isomerization of 2- to 1-butene are lower than previous estimates [6,13]; this is due to the calculated barrier for isomerization being  $\sim 15$ – $20$  kcal/mol higher in energy than previously estimated. The surface shown in Fig. 7 indicates that the dissociation of the 2-butene isomer may form allyl plus methyl by first isomerizing to 1-butene. Zhang et al. [12] proposed this reaction to account for allyl formation in their 2-butene pyrolysis experiments at  $\sim 0.01$  atm and the temperature range 900–1900 K; however, as seen in Fig. 8 the calculated rate constant in this study is  $\sim 2$ – $3$  orders of magnitude smaller than their estimate.

For 1-butene, the lowest energy decomposition pathway is the cleavage of a C–C bond to form allyl and methyl radical (R1).



The barrier for isomerization to 2-butene is 3 kcal/mol higher in energy than the C–C bond fission, and the pre-exponential factor is also much smaller ( $\sim 10^{15}$  versus  $\sim 10^{13}$  s $^{-1}$ ). The other bond fission pathways are at least 9 kcal/mol higher in energy. The barrier for molecular elimination to form 1,3-butadiene plus  $H_2$  is much higher.

For 2-butene, in contrast, the molecular elimination channel to form 1,3-butadiene plus  $H_2$  is the lowest energy pathway (R2).



2-Butene can also isomerize to 1-butene; however, the isomerization barrier is  $\sim 17$  kcal/mol higher (from the trans-isomer) than the direct elimination channel and therefore this pathway is insignificant (cf. Fig. S5 in the SI). The barriers for bond fission are much higher than the molecular channels, although the pre-exponential factors are also higher and may offset the high

barriers. The lowest energy bond fission reaction forms methyl allyl radical and H-atom (R3).

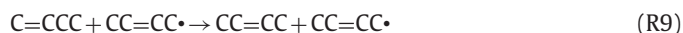


Even though the contribution from this pathway is small, compared to the molecular elimination channel (R2), it results in the formation of radicals that react with the fuel, accelerating its thermal decomposition. However, because the barrier for C-H bond cleavage is significantly higher than the C-C bond cleavage in 1-butene, 2-butene decomposes at higher temperatures than 1-butene (cf., Fig. 1).

Once the radical pool is generated, the fuel primarily decomposes via bimolecular abstraction and addition reactions. For 1-butene, both methyl and allyl radicals abstract an H-atom to form methyl allyl and methane or propylene (R4 and R5, respectively).



Methyl allyl undergoes a  $\beta$ -scission reaction to form H atoms and 1,3-butadiene (R6), and the H-atoms add to 1-butene (R7) to form propylene and methyl radical or abstract an H-atom to form methyl allyl plus H<sub>2</sub> (R8). Methyl allyl can also abstract an H-atom from 1-butene (R9), effectively converting 1-butene to 2-butene. Thus, under these conditions, 1-butene is converted to 2-butene in a bimolecular route rather a unimolecular route. This sequence of reactions forms a radical chain that drives the decomposition of 1-butene.



For 2-butene, in addition to the molecular H<sub>2</sub> elimination channel (R2), there is a significant contribution from the radical chain mechanism. Once formed, H-atoms add to 2-butene to form methyl radicals plus propylene (R10). Both H-atoms and methyl radicals can abstract a hydrogen atom from 2-butene to form methyl allyl and hydrogen or methane ((R11) and (R12), respectively). A subsequent  $\beta$ -scission reaction of methyl allyl forms 1,3-butadiene plus H-atom (R6), or methyl allyl can abstract from 2-butene (reverse of (R9)), effectively converting it to the more reactive 1-butene isomer.



A sensitivity analysis was performed to determine the impact of changing rate constant values on the concentrations of the measured species. Tables 1 and 2 display the sensitivity coefficients for the more important species for 1- and 2-butene pyrolysis, respectively. The full sets of results are provided in the SI (Tables S16 and S17 for 1-butene and Tables S18 and S19 for 2-butene). What is remarkable about this analysis is that, out of ~11000 reactions (when written irreversibly) in the mechanism, the predictions are sensitive to only a small number of reactions. For 1-butene pyrolysis, fuel decomposition is sensitive to four reactions; these are,

in order, (R1), (R7), (R4), and (R6). (Note that for two of these reactions, the reverse reaction is also important.) For 2-butene pyrolysis, eight reactions are sensitive; these are (R12), (R10), (R9), (R1), (R2), (R6), (R3), and (R11). The rate constants for most of these reactions are derived from electronic structure calculations. The H-abstraction reactions are based on rate estimation rules. In particular, (R5) and (R9) are near thermoneutral H-abstraction reactions in which a resonantly-stabilized radical abstracts from an olefin forming another stabilized radical and the conjugate olefin. These types of abstraction reactions have recently been investigated [25,32] and will be further discussed below. The reactions, (R1), (R2), (R3), (R6), (R7), and (R10), are pressure-dependent and their rate coefficients are calculated using a QRRK/MS analysis [48].

#### 4.2. Formation of light ( $\leq \text{C}_4$ ) products

The decomposition of 1- and 2-butene results in the formation of several light olefins. The predicted concentration profiles of some of these olefins are shown as a function of the reactor distance (and hence time) in Fig. 9 at 710 and 735 °C, respectively. For these conditions (~2.4 s and 50 mol% butene) the conversion of 1-butene is slightly higher than that of 2-butene (~96% versus 89%). While the two fuels produce the same light species, there are differences in the relative concentrations of these products. Most notable is that for 2-butene pyrolysis, the concentration of 1,3-butadiene is roughly twice that for 1-butene pyrolysis and it is the first olefin formed. In both cases, the concentration of 1,3-butadiene rises to a maxima then decreases at longer time. Propylene is formed more quickly in 1-butene pyrolysis and rises to a slightly higher mole fraction. Significantly more ethylene is produced in 1-butene pyrolysis.

Olefin formation occurs via the following reaction sequences. H-atom abstraction from the parent, 1- or 2-butene, leads to the formation of methyl allyl radical, and its subsequent  $\beta$ -scission reaction leads to the formation of 1,3-butadiene plus H-atom (R6). For 2-butene pyrolysis, 1,3-butadiene is also formed directly from molecular elimination of H<sub>2</sub> (R2); this channel accounts for its early formation and the higher mole fraction in 2-butene pyrolysis. The primary formation pathways for propylene are H-atom addition to the fuel ((R7) and (R10)) or H-atom abstraction from the fuel by allyl radical ((R5) and (R13)). The addition reaction of H-atom to propylene forms ethylene plus methyl radical (R14). Two other minor ethylene formation pathways are  $\beta$ -scission reaction of methyl allyl (R15) or H-atom addition to 1,3-butadiene (R16).



The subsequent H-atom abstraction reactions from the stable species by vinyl radical are highly exothermic. These reactions form more ethylene, and they can result in the formation of other vinylic radicals such as  $\text{C}=\text{C}\cdot\text{CC}$  (from 1-butene) and  $\text{CC}=\text{C}\cdot\text{C}$  (from 2-butene), which decompose to form allene and propyne plus methyl ((R17) and (R18)), respectively. Allene is also produced from the disproportionation reaction between allyl and 1-methyl-allyl (R19), or allyl and other resonantly stabilized radicals. Allene and propyne can interconvert via a cyclic C<sub>3</sub>H<sub>4</sub> species [49] or with the assistance of H atoms [50].







Normalized sensitivity coefficients calculated at 735 °C for the major species in 2-butene pyrolysis.

[illegible]

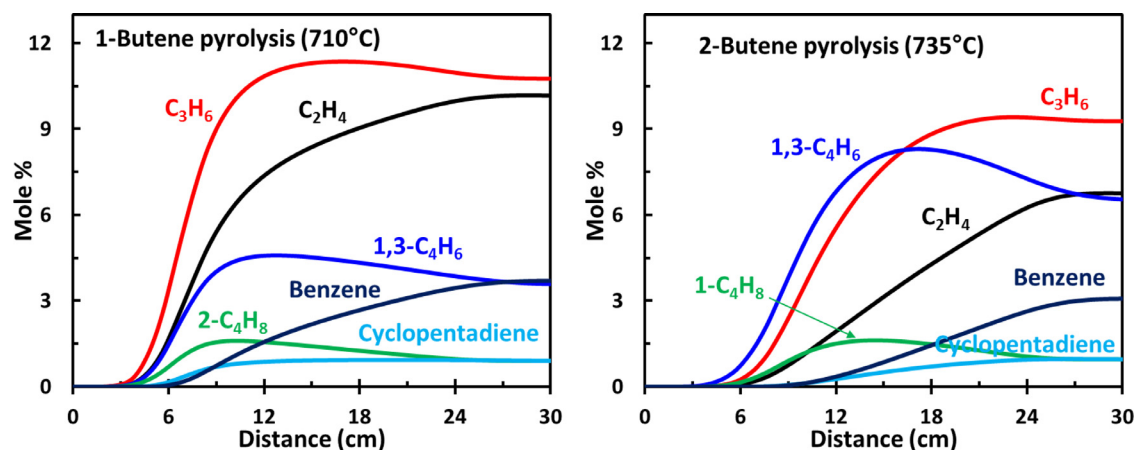
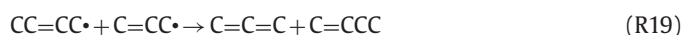


Fig. 9. Predicted formation of ethylene, propene, 1- or 2-butene, 1,3-butadiene, 1,3-cyclopentadiene, and benzene as a function of reactor distance during the pyrolysis of 1-butene at 710 °C (left) and 2-butene at 735 °C (right). The initial fuel input is 50 mol% and the residence time is ~2.4s.



For both butenes, H-atom abstraction from the fuel by methyl allyl radical leads to the formation of the other butene isomer (R9). Hydrogen and the light alkanes are formed from H-atom abstraction reactions by their respective radicals, primarily from the parent fuel. Ethyl, propyl, and butyl radicals are formed primarily from H-atom addition to the corresponding olefin or from the  $\beta$ -scission reactions of larger alkyl radicals.

The sensitivity coefficients for the most sensitive reactions that produce the light products are shown in Tables 1 and 2. Once again we see that most species are sensitive to only a few reactions. Furthermore, out of the sensitive reactions, only a few reactions have a very high sensitivity coefficient ( $>0.5$ ). In many cases, the same set of reactions that impact the fuel decomposition also impact the formation of several light products. Most of the other reactions have been identified above. The relatively small number of sensitive reactions has some important implications for the mechanism development process. As long as the rate parameter assignments for the various reactions are reasonably close to the actual values, the mechanism predictions should be fairly reliable. At that point, a sensitivity analysis can be used to identify those reactions that may warrant further investigation. In contrast, if every reaction in the mechanism warranted such detailed attention, the mechanism development process would become too cumbersome for larger sized fuels.

#### 4.3. Radical pool distribution

One characteristic of olefin pyrolysis is that it leads to the formation of high concentrations of resonantly stabilized radicals. The predicted concentration profiles of several such radicals are compared with the non-stabilized radicals in Fig. 10 for 1-butene pyrolysis at 710 °C and 2-butene pyrolysis at 735 °C. In both cases, the predicted concentrations of methyl allyl, allyl, and cyclopentadienyl radical are at least an order of magnitude greater than that of methyl, and three to four orders of magnitude greater than that of vinyl and H-atom. The consequence of this is that, even though the rate constants for their H-abstraction and addition reactions are small, the rates of these reactions are potentially significant and can lead to the formation of MWG products [25,26]. Recombination reactions of resonantly stabilized radicals provide an additional MWG pathway. Comparison of the radical distribution produced by the two fuels shows that there are some differ-

ences in the times (and hence distances) when the various radicals are formed. For instance, allyl, methyl allyl, methyl, and ethyl are formed very early for 1-butene pyrolysis, while for 2-butene pyrolysis, methyl allyl is the only radical formed at early times. In the center of the reactor the concentrations of the various radicals are similar for the two fuels. However, since the radical pool is formed earlier for 1-butene pyrolysis, the total concentrations of several radicals are higher than that for 2-butene pyrolysis.

#### 4.4. Formation of MWG species

In this section we discuss some of the important reaction pathways that lead to the formation of 1,3-cyclopentadiene, benzene, and toluene. Benzene is the dominant MWG product; cyclopentadiene production is approximately three times smaller and is comparable to toluene production. As shown earlier, the model predictions for these three species are quite good. Analysis of the reaction mechanism reveals that there are multiple pathways to the formation of these products. However, we focus on radical addition reactions that are the primary pathways for production of MWG species. Radical addition reactions to olefins and diolefins have been shown to be an important route to MWG species formation [25]. Here we discuss four reactions of this type: (1) vinyl addition to 1,3-butadiene, (2) allyl addition to 1,3-butadiene, (3) allyl addition to 1-butene, and (4) allyl addition to 2-butene.

An important feature of 1- and 2-butene pyrolysis is that 1,3-butadiene is formed in relatively high concentrations. This is significant because radical addition to the terminal carbon atom in 1,3-butadiene results in formation of a resonantly stabilized adduct. This added stability lowers the energy barrier for formation of cyclic species relative to redissociation back to reactants. An extreme case is the addition of vinyl radical to 1,3-butadiene, which combines the added stability of the adduct with a very high energy entrance channel. A detailed analysis of the C<sub>6</sub>H<sub>9</sub> PES has recently been provided [25,51] and this PES is provided in the SI (Fig. S7); only the lowest energy channels are shown in Fig. 11. Vinyl addition to the terminal end of 1,3-butadiene forms the resonantly stabilized adduct, 1,5-hexadiene-3-yl radical (1) with a barrier of only ~2.3 kcal/mol. The barriers to form either five- or six-member cyclic species are much lower in energy than redissociation; as a result, these pathways are favorable even though the pre-exponential factors for the isomerizations are lower than that for redissociation. There are two primary pathways that lead to the formation of the five-membered ring products. Exo-cyclization leads to the formation of the cyclopentene-4-carbinyl radical (4). The cyclopentene-4-carbinyl radical (4) can then undergo

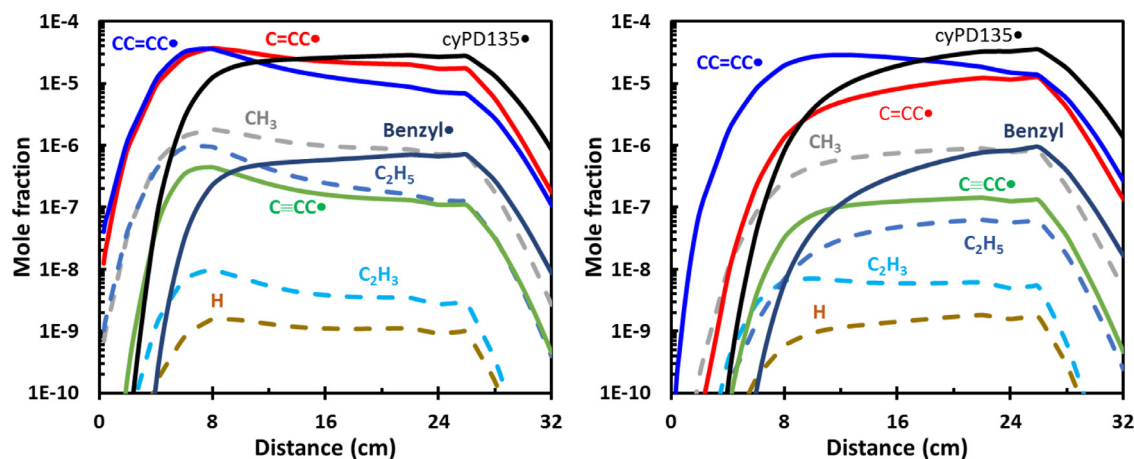


Fig. 10. Predicted mole fractions as a function of distance for selected radicals that are formed during the pyrolysis of 1-butene at 710 °C (left) and 2-butene at 735 °C (right). The initial fuel input is 50 mol% and the residence time is  $\sim 2.4$  s. Note the cyPD135• denotes the 1,3-cyclopentadienyl radical.

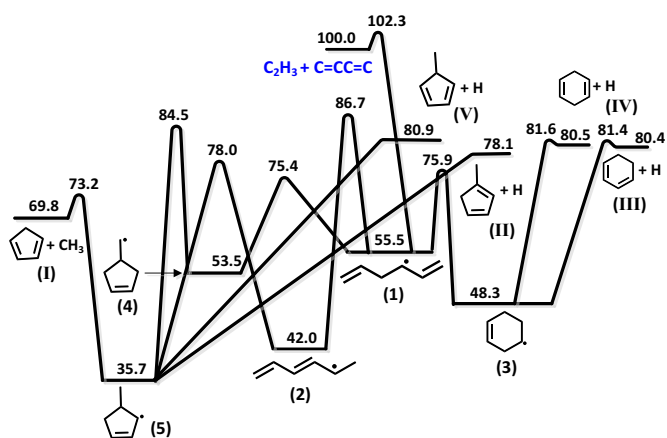


Fig. 11. Simplified  $C_6H_9$  PES calculated at the CBS-QB3 level of theory. For simplicity only one resonance structure is shown for (1), (2) and (5). Enthalpies are provided at 298 K in kcal/mol.

an intramolecular H-atom shift reaction via a four-membered ring to 4-methyl-cyclopentene-3-yl radical (5) and subsequent  $\beta$ -scission reactions results in the formation of cyclopentadiene and methyl (I) or methyl cyclopentadiene isomers and H-atoms (II and V). Alternatively, (1) undergoes an H-atom shift to form 1,3-hexadien-5-yl (2), which then forms (5) via an endo-cyclization; the 4-methyl-cyclopentene-3-yl radical (5) may then react as described above. Stable six-membered ring products may be formed by endo-cyclization of (1) to form the cyclohexen-4-yl radical (3). A subsequent  $\beta$ -scission leads to the formation of the 1,4- and 1,3-cyclohexadiene isomers plus H-atom (III and IV).

The addition reaction of allyl to 1,3-butadiene via the  $C_7H_{11}$  PES provides another route to the formation of stable MWG species. This surface has previously been discussed in the context of MWG species formation [25] and is included as Fig. S8 in the SI; the lowest energy pathways are shown in Fig. 12. This reaction is an interesting contrast to the vinyl addition reaction in that the barrier for addition is higher ( $\sim 7.5$  versus 2.3 kcal/mol) and the barrier for redissociation is lower ( $\sim 26$  versus  $\sim 47$  kcal/mol). Nevertheless, since the concentration of allyl radical is higher than vinyl by approximately three orders of magnitude (cf. Fig. 10); allyl addition is important. Addition to the terminal end of 1,3-butadiene results in the formation of 1,6-heptadiene-3-yl radical (1). Even with the shallower well for this case, several exit chan-

nels have barriers that are lower in energy than that for redissociation out the entrance channel. One expects redissociation will be more significant for this case since the pre-exponential factor for formation of the isomers via cyclic transition states will be much lower. Once formed, the 1,6-heptadiene-3-yl radical (1) can undergo a H-atom shift reaction via six-membered ring transition state to form the 1,5-heptadiene-3-yl isomer (3), which then may cyclize via an endo-addition to form 4-methyl-cyclohexene-5-yl radical (4). A subsequent  $\beta$ -scission reaction of (4) leads to the formation 1,4-cyclohexene and methyl (I) and, to a lesser extent, methyl-cyclohexene isomers and H-atoms (not shown). The 1,6-heptadiene-3-yl radical (1) can also directly cyclize via an exo-addition to form the cyclohexene-4-carbinyl radical (5), which can  $\beta$ -scission to form 4-vinyl-cyclohexene and H (III) or ring open to form 2,4-hexadiene-6-yl radical (6). Cyclization of (6) leads to the formation of the 1,4-dimethyl-cyclopentene-5-yl radical (7), which can  $\beta$ -scission to form 5-methyl 1,3-cyclopentadiene and methyl (II). The barriers for the pathways from (5) to (III) or to (6) are slightly above that of the entrance channel.

Radical addition reactions to olefins may also be important. However, here the initial adduct is not resonantly stabilized, and the likelihood that the adduct will simply dissociate back to the reactants is greater than in the above examples. Nevertheless, these reactions need to be considered since the concentration of olefins are larger than that of the 1,3-butadiene. This is especially the case at early stages of the reaction where the concentration of the parent olefin is high. The addition reaction of allyl radical to 1- or 2-butene occurs on the  $C_7H_{13}$  PES; for clarity, these two addition channels are shown separately in Figs. 13 and 14, respectively. The two channels are connected via an energetically unfavorable H-atom shift reaction via 4-member transition state ring (not shown). The energetics of the two addition pathways are similar. Allyl addition to 1-butene (Fig. 13) results in the formation of the 4-ethyl-1-pentene-5-yl radical (1) or the 1-heptene-5-yl radical (2). Allyl addition to 2-butene (Fig. 14) results in the formation of 4-methyl-1-hexene-1-yl radical (1). Compared to allyl addition to 1,3-butadiene (in Fig. 12), the addition barriers are  $\sim 6$ – $7.5$  kcal/mol higher, while the redissociation barriers are  $\sim 4$  kcal/mol lower in energy. The higher addition barrier leads to a smaller addition rate constant, while the lower redissociation barrier decreases the likelihood that MWG species will be formed. However, there are still multiple MWG pathways that have lower energy barriers than that of the entrance channel. Cyclization of the initial adduct through both an endo- or exo-addition is energetically favored and the subsequent reactions eventually generate stable five-membered ring

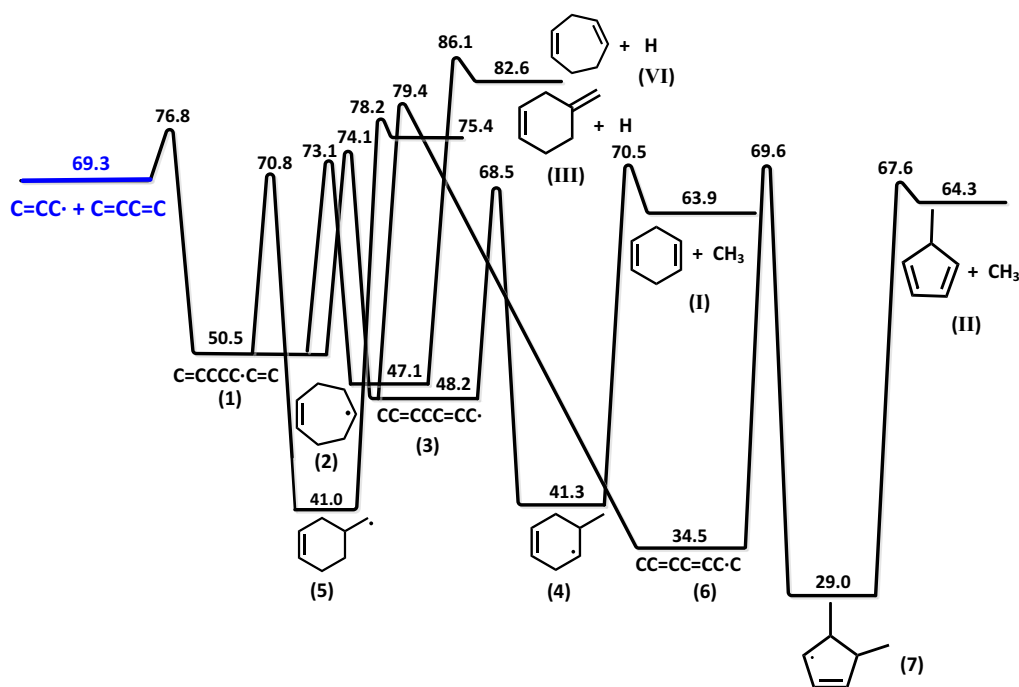


Fig. 12. Simplified  $C_7H_{11}$  PES calculated at the CBS-QB3 level of theory. For simplicity only one resonant form is shown for (1), (3), (6), and (7). Enthalpies are provided at 298 K in kcal/mol.

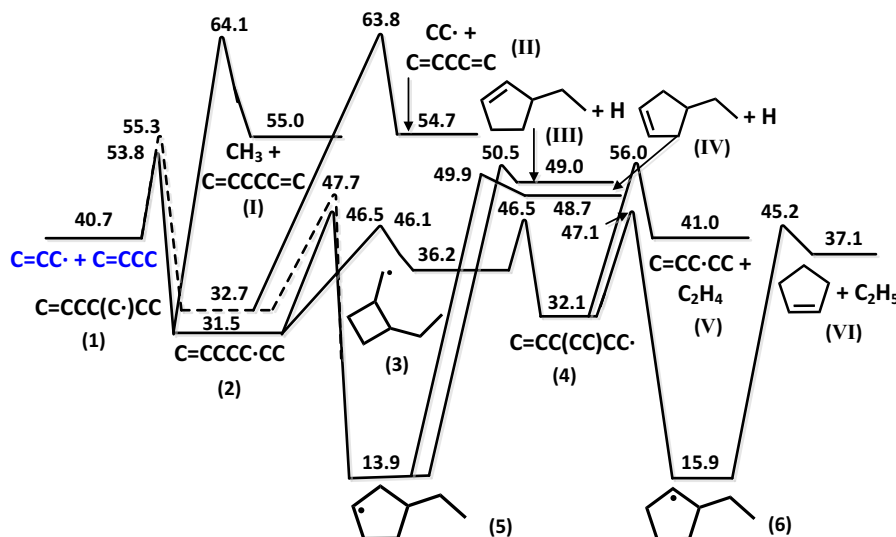


Fig. 13. Simplified  $C_7H_{13}$  PES calculated at the CBS-QB3 level of theory for the terminal (solid line) and non-terminal addition (dashed line) of allyl to 1-butene. Enthalpies are provided at 298 K in kcal/mol.

species (mostly (III) and (IV) as well as (VI) in Fig. 13 and mostly (IV) as well as (V) in Fig. 14). The apparent rate constants at 1 atm forming the MWG species from the  $C_7H_{13}$  PES are shown in Fig. S9 in the SI.

The four addition pathways discussed above are primary routes to the formation of stable five- and six-membered ring species. Three other reactions that appear to be important are the additions of methyl allyl to 1,3-butadiene (see Fig. S10 in the SI) and to 1- and 2-butene. These reactions are not discussed in detail because they proceed analogously to that of allyl addition to 1,3-butadiene and 1- and 2-butene, respectively. The addition reactions of methyl allyl and allyl to acetylene, ethylene, and propylene also contribute to MWG formation. The reactions of allyl to these three species (the  $C_5H_7$ ,  $C_5H_9$ , and  $C_6H_{11}$  PESs) have previously been discussed [25,26].

Figure 15 shows the rates of production for 1,3-cyclopentadiene during 1-butene pyrolysis (left plot). The total rate of production is shown by the black dashed line. The most important reaction producing cyclopentadiene is the  $\beta$ -scission reaction of the 4-ethyl-cyclopentene-3-yl radical (red line in figure). This species is formed from abstraction from 3- or 4-ethyl-cyclopentene, which result from the addition reactions of allyl and methyl allyl to 1-butene (cf. Fig. 13). This reaction sequence occurs quite early since these two radicals are formed very early (cf. Fig. 10) and the concentration of the fuel is also high at early reaction times. As the reactions proceed, vinyl radical addition to the 1,3-butadiene (blue line) becomes an important pathway for cyclopentadiene formation (cf. Fig. 11). The addition reaction of allyl to acetylene, shown in green, is a minor pathway. Although the acetylene concentration is very low (typically predicted to be less than 2% of the ethylene



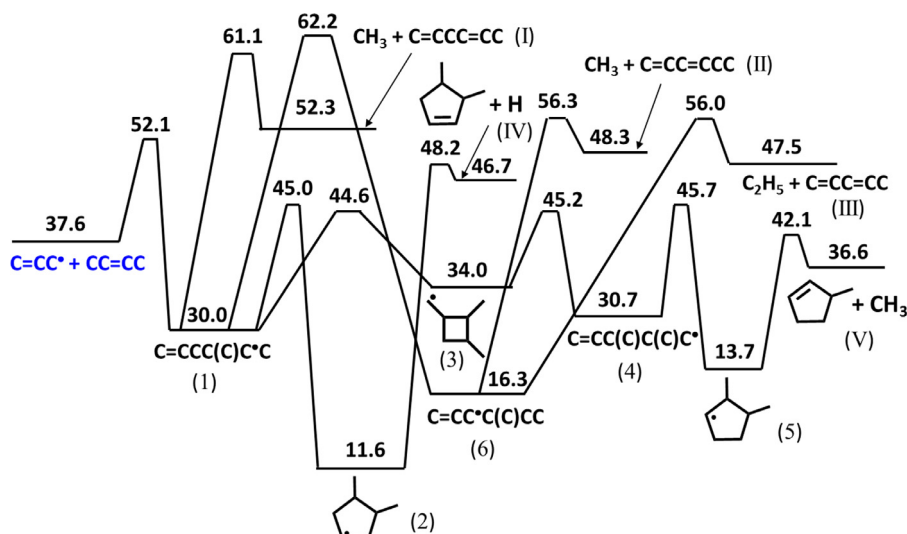


Fig. 14. Simplified  $C_7H_{13}$  PES calculated at the CBS-QB3 level of theory for the addition of allyl to 2-butene. Enthalpies are provided at 298 K in kcal/mol.

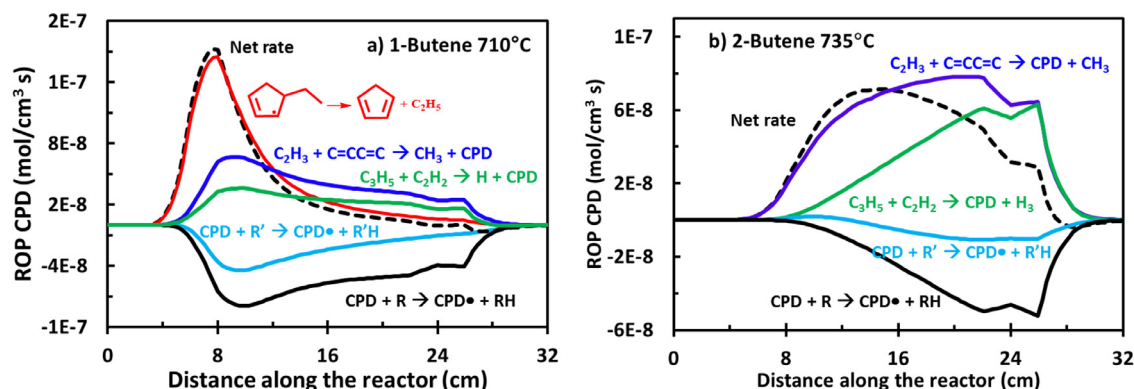


Fig. 15. Rate of production plots for 1,3-cyclopentadiene (CPD) formed during 1-butene pyrolysis at 710 °C (left) and 2-butene pyrolysis at 735 °C (right). The initial fuel input is 50 mol% and the residence time is 2.4 sec. The primary decomposition pathways are H-atom abstraction by normal (non-resonantly stabilized) radicals (R) and resonantly stabilized radicals (R').

concentration), the presence of low energy pathways to cyclopentadiene leads to quite large rate constants [26]. Cyclopentadiene is consumed by a variety of H-atom abstraction reactions. An interesting feature is that the weak bond in cyclopentadiene allows for abstraction by resonantly stabilized radicals (R', shown by the light blue line) as well as by H, CH<sub>3</sub>, and other non-resonantly stabilized radicals (R, shown by the black line). The significance of the near thermoneutral abstraction reactions is discussed in detail below. Although the rate constants are much lower than those for the highly exothermic abstractions by H and CH<sub>3</sub>, the much larger concentrations of the resonantly stabilized radicals make their abstractions competitive.

The right panel in Fig. 15 shows the production rates of 1,3-cyclopentadiene for 2-butene pyrolysis. Here 1,3-cyclopentadiene is formed later in the reaction. In this case, the reaction of allyl and methyl allyl to 2-butene results in the formation of methyl cyclopentadiene rather than 1,3-cyclopentadiene. The dominant pathway to 1,3-cyclopentadiene is vinyl addition to 1,3-butadiene with a minor contribution from allyl addition to acetylene. Abstraction reactions of 1,3-cyclopentadiene are slightly less important than is the case for 1-butene since it is produced nearer the reactor exit, reducing the time available for such reactions.

The rates of production plots for benzene are shown in Fig. 16. The left panel has the results for 1-butene pyrolysis and the right those for 2-butene pyrolysis. For both fuels, the primary formation pathway is from  $\beta$ -scission of an H-atom from the

1,3-cyclohexadienyl radical, shown by the red lines, and to a lesser extent from  $\beta$ -scission of CH<sub>3</sub> from methyl 1,3-cyclohexadienyl, shown by the blue lines. These two radicals are formed from H-atom abstraction from their parent species, which are experimentally detected (cf. Figs. 3 and 4). The parent species result from the reactions of vinyl (Fig. 11), allyl (Fig. 12), and/or methyl allyl (Fig. S10) addition to 1,3-butadiene. For 1-butene pyrolysis the allyl addition reaction is favored, while for 2-butene pyrolysis the vinyl reaction is favored. The other significant formation channel, shown by the green lines, is the isomerization of the methyl-cyclopentadienyl radicals to form benzene plus H. This reaction occurs on the C<sub>6</sub>H<sub>7</sub> surface, which has previously been discussed [26].

The rates of production analysis for the toluene formation in 1- and 2-butene pyrolysis at 710 °C and 735 °C are shown in the SI (Fig. S11). As shown in these plots, the major pathways are the same for both fuels, except that the net rate for toluene formation is slightly faster for 1-butene than that of 2-butene and its formation occurs earlier on in the reactor. In both cases, toluene is primarily formed from the  $\beta$ -scission of an H-atom from methyl-1,3-cyclohexadienyl radical. This radical is formed from the addition of methyl allyl to 1,3-butadiene (see Fig. S10).

The predictions indicate that these three major MWG species are formed earlier in 1-butene pyrolysis than in 2-butene pyrolysis (cf. Fig. 9). For 1,3-cyclopentadiene two factors contribute to this difference. The first is the more rapid production of allyl and

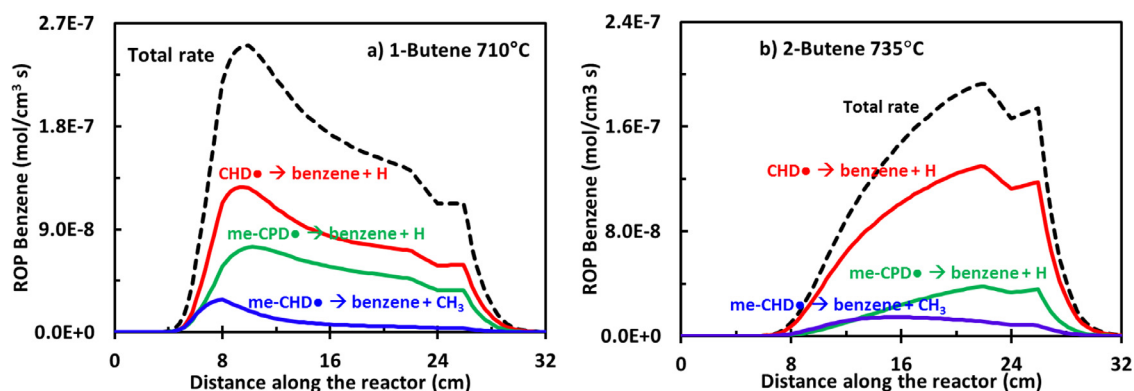


Fig. 16. Rate of production plots for benzene formed during 1-butene pyrolysis at 710 °C (left) and 2-butene pyrolysis at 735 °C (right). The initial fuel input is 50 mol% and the residence time is 2.4 s. (CHD• is the cyclohexadienyl radicals; me-CHD• is methyl cyclohexadienyl radicals, and me-CPD• are methyl cyclopentadienyl radicals.)

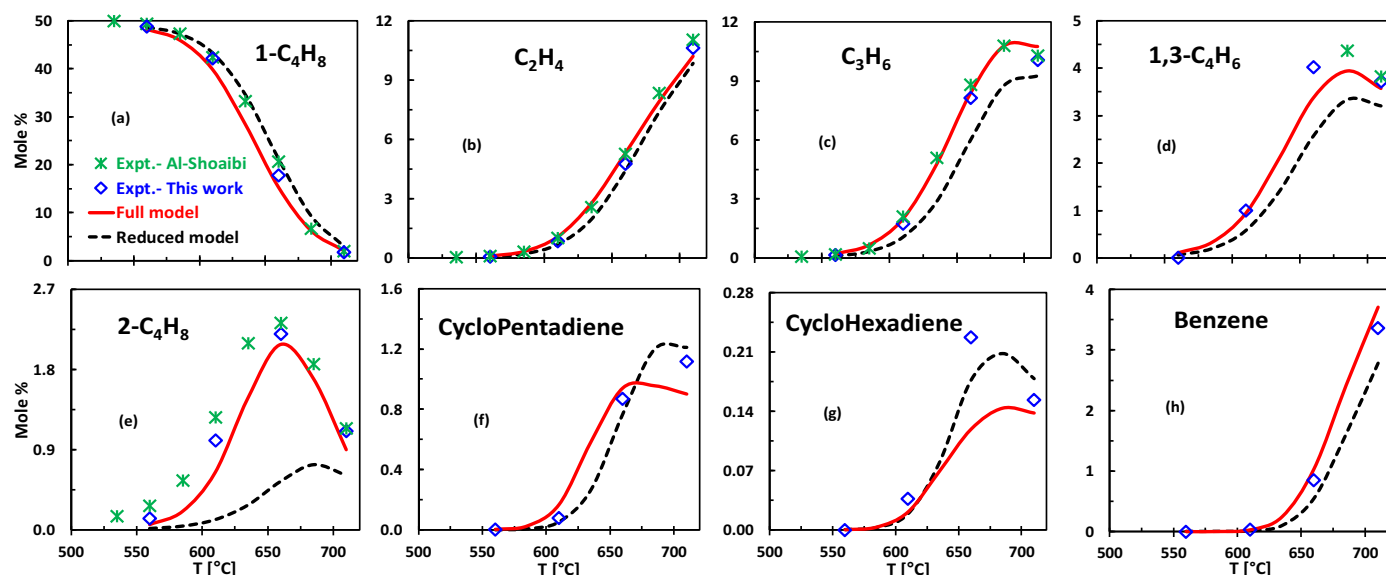


Fig. 17. Comparison of observed mole fractions to modeling predictions for the pyrolysis of 1-butene at a residence time of 2.4 s. The initial fuel input is 50 mol%. Note that the “reduced” model excludes all the “thermoneutral” hydrogen abstraction reactions. (Symbols – experimental data; red solid lines – full model, black dashed lines – “reduced” model.)

methyl allyl in 1-butene pyrolysis. The second is that for 1-butene pyrolysis, addition to the parent fuel is important, while for 2-butene pyrolysis the primary addition pathway is to the product 1,3-butadiene. In the case of benzene and toluene, the main difference is the rapid production of allyl and methyl allyl in 1-butene pyrolysis.

#### 4.5. Impact of thermoneutral H-abstraction reactions

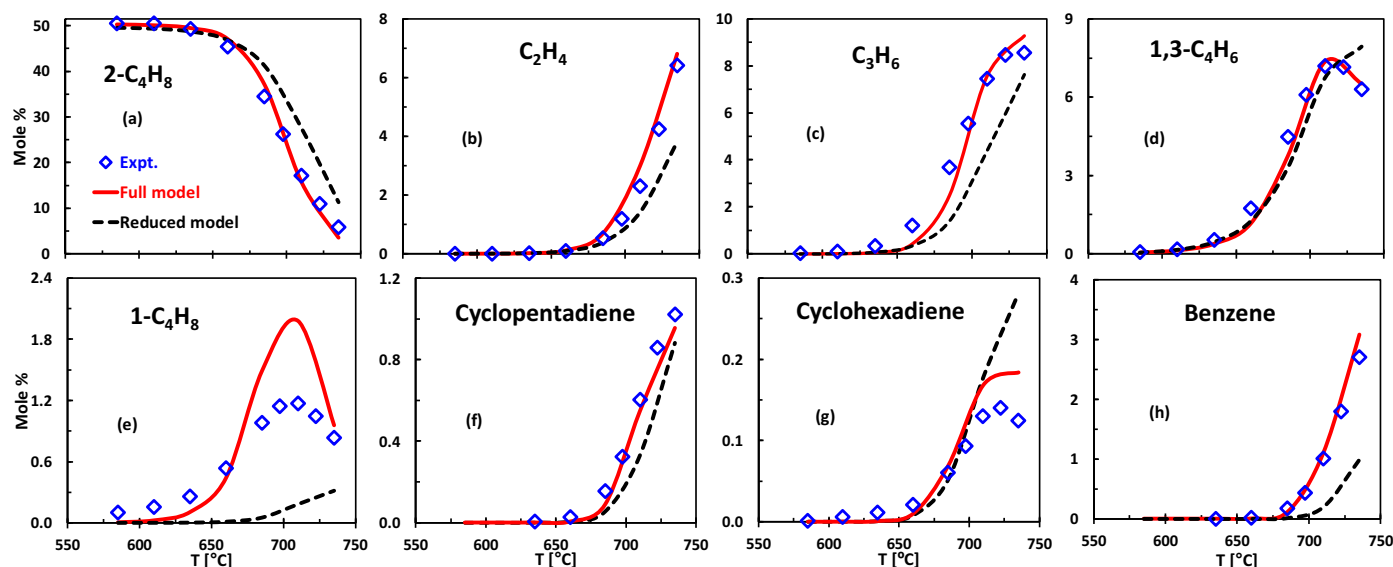
The observed product distributions are also impacted by the approximately thermoneutral H-atom abstraction reactions involving resonantly stabilized radicals (e.g., (R5) and (R9)). One consequence of this is that these reactions may shift the distribution of resonantly stabilized radicals that is initially formed. Given the importance of these species in terms of the MWG kinetics, this redistribution has the potential to shift the product distribution. A detailed analysis of the rate constants for these reactions has previously been provided. [25]

To test the impact of these reactions, we compare the predictions of the model with that of a reduced model, where all approximately thermoneutral H-abstraction reactions are excluded from the full mechanism. The predictions for the two models are shown for several selected species in Figs. 17 and 18 along with the data

for 1- and 2-butene pyrolysis, respectively. The trends are the same for both fuels. The reduced model underpredicts the fuel conversion, especially for 2-butene pyrolysis. Propylene, 1,3-butadiene, and either 1- or 2-butene are also underpredicted for both species. These differences are primarily due to the exclusion of the reactions of allyl and methyl allyl abstraction from the parent fuel. A consequence of the absence of the allyl abstraction reaction is that the concentration of methyl allyl is lower, which results in a lower concentration of 1,3-butadiene. The concentration of propylene is also lower, and therefore less ethylene is formed. The H-atom abstraction reaction of cyclopentadienyl radical with the parent fuel and other olefins are excluded. This leads to a lower concentration of 1,3-cyclopentadiene under most conditions. The abstraction reactions of allyl and methyl allyl with cyclohexadiene are also excluded. As a result, even though the concentration of 1,3-butadiene is lower, cyclohexadiene is overpredicted and therefore benzene is underpredicted.

#### 4.6. Comparisons to other studies

Comparisons to literature data allow us to test the robustness of the kinetic model over an extended temperature and pressure



**Fig. 18.** Comparison of observed mole fractions to modeling predictions for the pyrolysis of 2-butene at a residence time of 2.4 s. The initial fuel input is 50 mol%. Note that the “reduced” model excludes all the “thermoneutral” hydrogen abstraction reactions. (Symbols – experimental data; red solid lines – full model, black dashed lines – “reduced” model.)

range. It also allows us to illustrate the importance of the pyrolysis reaction subset for oxidation/combustion systems.

*Low pressure, high temperature pyrolysis of 1- and 2-butene.* Recognizing that so many of the reactions in the model involved chemically-activated reactions, it is essential that the model properly account for the variation in the branching ratios with both temperature and pressure. We can assess these predictions by comparison to data taken under much different conditions. Zhang et al. [12] reported a pyrolysis experiment for these butene isomers at low pressures ( $\sim 3$ –11 Torr) in a flow reactor where 4 mol% of the fuel was diluted in argon. The pyrolysis temperature ranged from 900 to 1900 K with residence times of 1.25–1.33 ms. Several isomeric products and intermediate radicals were identified using a synchrotron vacuum ultraviolet (VUV) photoionization mass spectrometry. The authors observed good agreement when the data was compared to the predictions of a small kinetic mechanisms (76 species) that was based on the USC Mech II model [2]. Figure 19 compares the predictions using the model developed in this study to the 1-butene pyrolysis experiment by Zhang et al. [12]. The model accurately captures the fuel conversion and most of the stable species profiles: methane, ethane, ethylene, acetylene, propylene, allene, vinyl acetylene, diacetylene, 1,3-cyclopentadiene, and benzene. The concentrations of hydrogen and 1,3-butadiene are slightly underpredicted, while propyne is overpredicted. The temperature-dependent predictions for the radicals, methyl, allyl, and propargyl, are encouraging. The predictions for methyl and propargyl may well be within the uncertainty of the measurements, although the allyl predictions appear to be low.

The analogous 2-butene pyrolysis comparisons are shown in Fig. 20. The fuel conversion is well predicted, but overall the comparison of the product distributions are not as good as for the 1-butene experiments. A few are quite good: hydrogen, methane, propylene, vinylacetylene, diacetylene, and benzene. Acetylene, ethylene, and 1,3-butadiene are overpredicted, while propyne and cyclopentadiene are underpredicted. The profiles for the radicals, methyl and propargyl, are really quite good, but allyl is again underpredicted. It is interesting to note that in some cases the observed deviations in the model are opposite for the two fuels (e.g., propyne). Overall, the results are encouraging, especially considering the substantially lower pressure, significantly higher tempera-

tures, and much faster residence times than in our experiments. Further improvements to the model can likely be made by examining the higher temperature kinetics.

*Oxidation of 1- and 2-butene.* As mentioned above, depending upon the conditions, an improved understanding of pyrolysis kinetics may also be important for development of improved oxidation models. To illustrate this, we compare the predictions of the pyrolysis model developed here to the data collected for the oxidation of 1- and 2-butene. The data is from two studies by Fenard et al. [14,15] that investigated the oxidation of these fuels (1-, cis-2-, and trans-2-butene) in a jet-stirred reactor at atmospheric pressure, with a residence time of 70 msec, and over a temperature range of 900 – 1250 K. Figs. 21 and 22 show both the oxidation data for an equivalence ratio of 0.5 along with the pyrolysis model predictions. The vertical lines, marking the first appearance of CO, designate the approximate temperature where the transition from pyrolysis to oxidation occurs. The predictions for the butene decay and product formation are quite good at lower temperatures ( $\leq 1075$  K for 1-butene and  $\leq 1100$  K for 2-butene). Note in Fig. 22, the oxidation of cis- and trans-2-butene are identical. At higher temperatures, oxidation products appear, and the model fails to predict the observed product distributions, as expected. Nevertheless, this comparison shows that, under certain conditions, the pyrolysis chemistry can influence the initial product distribution that then goes on to react via the oxidation chemistry. Thus, under these conditions, it is essential to have an accurate description of the pyrolysis chemistry in order to accurately predict the oxidation kinetics. Alternatively, it suggests that it could be possible to estimate the ignition behavior of a given fuel by comparing the predicted pyrolysis products to those of fuels where the ignition behavior is known.

## 5. Summary

In this work, we present a fundamentally-based detailed kinetic model that can accurately describe the pyrolysis of the 1- and 2-butene. The model predictions are compared to atmospheric (high-altitude) pressure flow reactor experiments that varied the initial fuel composition, residence time, and temperature. In some of these experiments, the fuel conversion was high and a

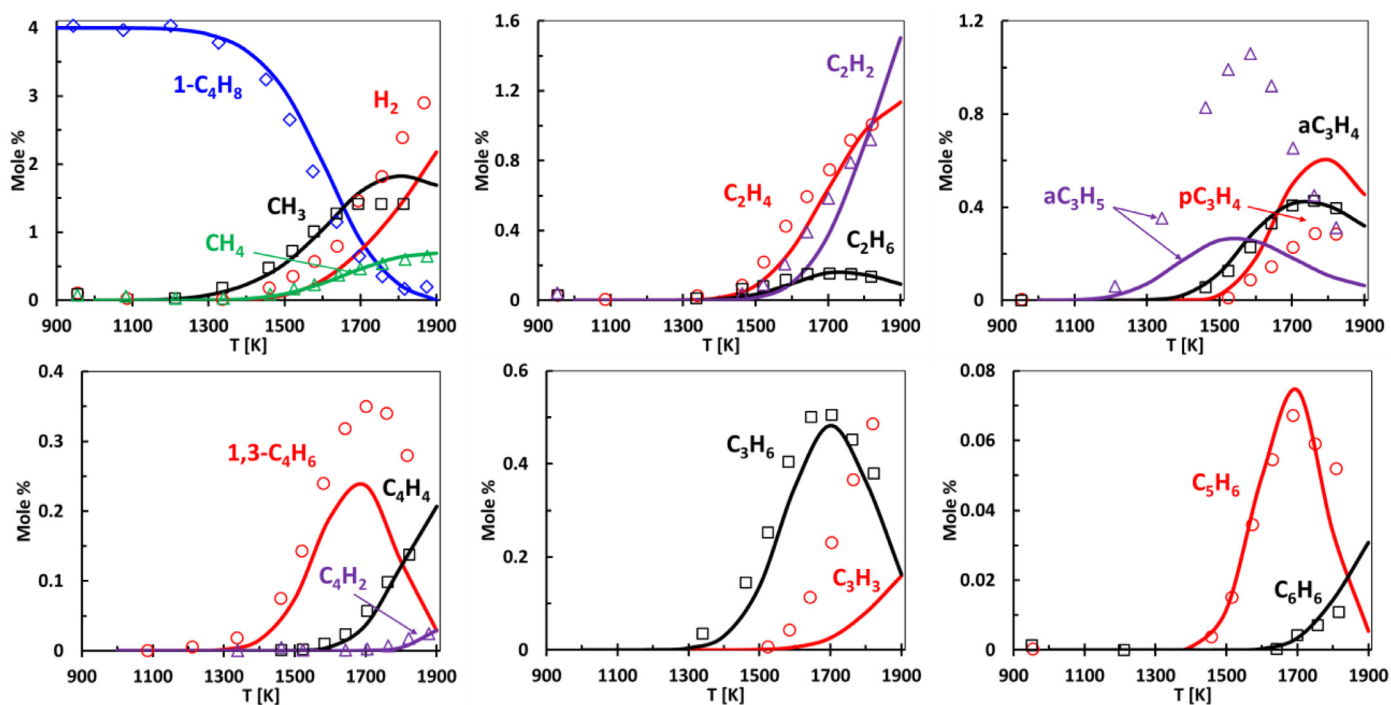


Fig. 19. Comparison of the model predictions (lines) to the literature data [12] (symbols) of 1-butene pyrolysis (initial conditions: 4% fuel in argon;  $p = 3\text{--}11$  Torr;  $\tau = 1.25\text{--}1.33$  ms).

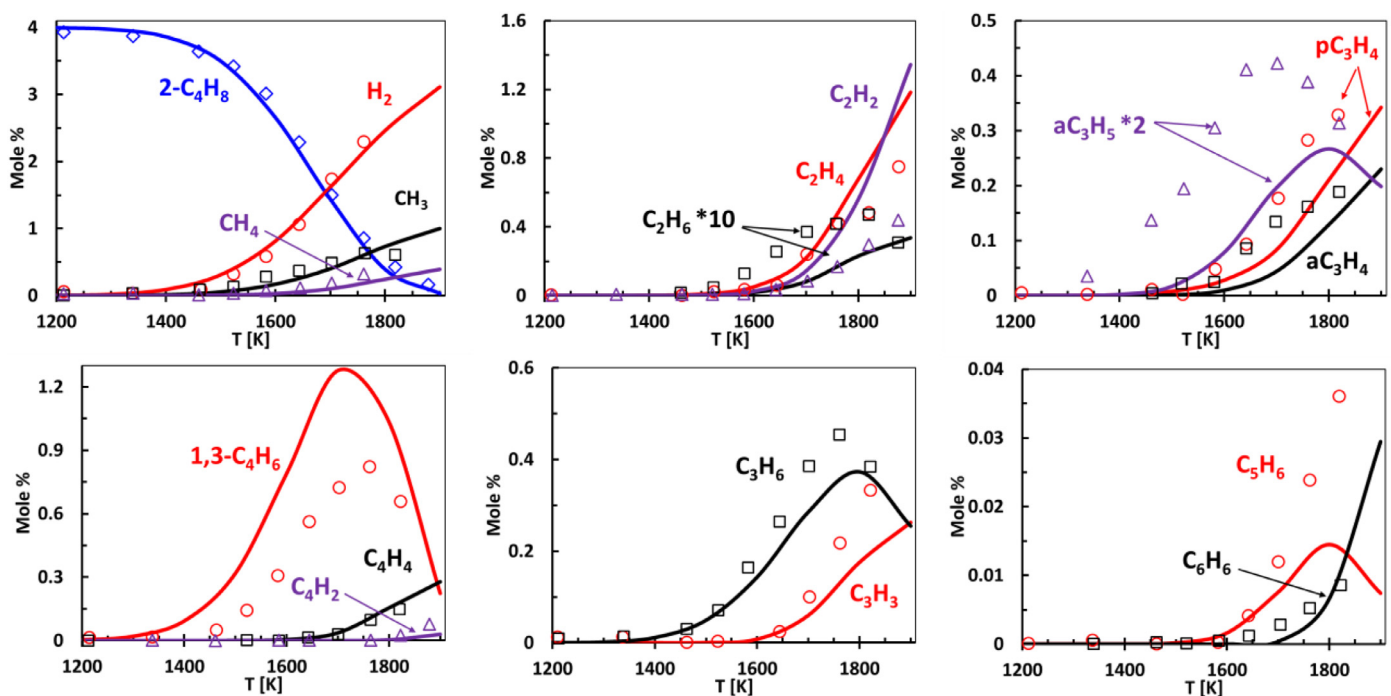
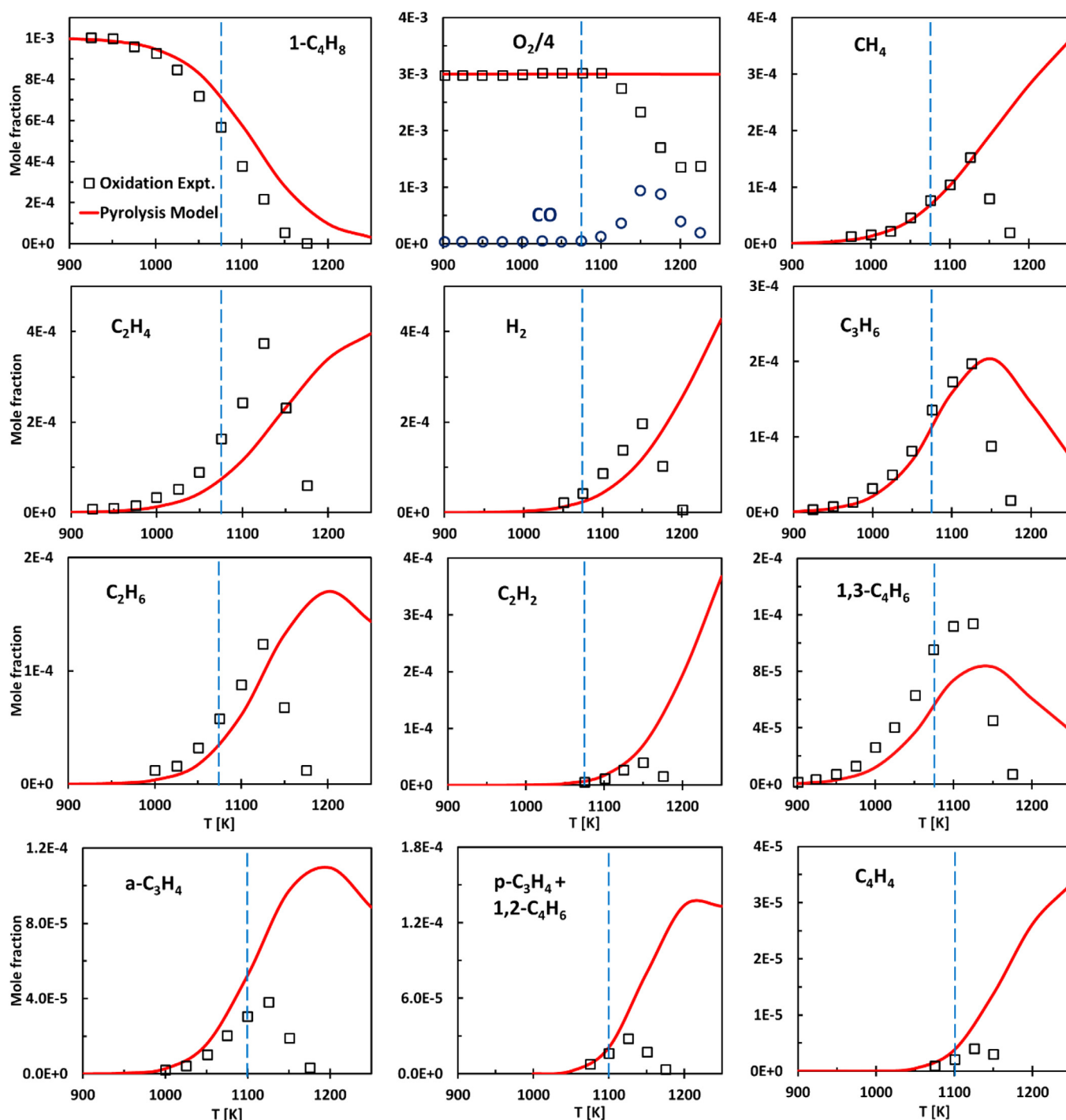


Fig. 20. Comparison of the model predictions (lines) to the literature data [12] (symbols) of 2-butene pyrolysis (initial conditions: 4% fuel in argon;  $p = 3\text{--}11$  Torr;  $\tau = 1.25\text{--}1.33$  ms).

variety of molecular weight growth (MWG) products were quantitatively identified. Benzene, 1,3-cyclopentadiene, and toluene were the most prevalent MWG products. The model is able to accurately predict the fuel conversions for both fuels as well as the formation of light species and major MWG products. The model was also compared to previously published pyrolysis data that was collected in a flow reactor at significantly shorter residence time, much lower pressure, and higher temperatures. Under these

conditions the agreement between the model and the data is satisfactory. The model was also applied to relatively low temperature oxidation data to illustrate the importance of accurately characterizing the pyrolysis kinetics.

The primary pathways that lead to the fuel decomposition and formation of the major products in experiments performed in this work are discussed. 1-Butene pyrolysis is initiated by cleavage of the weak allylic C–C bond to form allyl and methyl radicals. These



**Fig. 21.** Comparison of the model predictions (lines) to the literature data (symbols) of 1-butene oxidation in a jet stirred reactor at  $\Phi = 0.5$ ,  $p = 1$  atm,  $\tau = 70$  ms. [15]. The vertical line marks the temperature at which CO production becomes apparent, marking the transition from pyrolysis to oxidation kinetics.

radicals can abstract from the parent to form methyl allyl radical. For 2-butene pyrolysis, the fuel can directly decompose to 1,3-butadiene plus hydrogen. Cleavage of the allylic C-H bond to form methyl allyl and H-atom occurs in parallel. In both cases, the decomposition of the fuel leads to formation of a significant amount of 1,3-butadiene. This is significant because radical addition to the terminal carbon atom in 1,3-butadiene results in a resonantly stabilized adduct, and the added stability of this adduct lowers the energy required for the formation of stable five- and six-membered ring species. As specific examples of this, we discuss the addition reactions of vinyl and allyl addition to 1,3-butadiene.

Resonant radical addition reactions to olefins are also important. In particular, in the early stages of the reaction, radical addition to the parent olefin is important since its concentration is high even though the adduct well is relatively shallow. As an example of this, we discuss allyl addition to 1- and 2-butene. Lastly, we illustrate the importance of near thermoneutral resonant radical H-atom abstraction reactions from olefins. Even though the rate constants for these reactions are lower than those for the analogous reactions involving alkyl radicals, the higher concentration of resonantly stabilized radicals means the rates of these reactions may be significant. The occurrence of these reactions changes the resonant



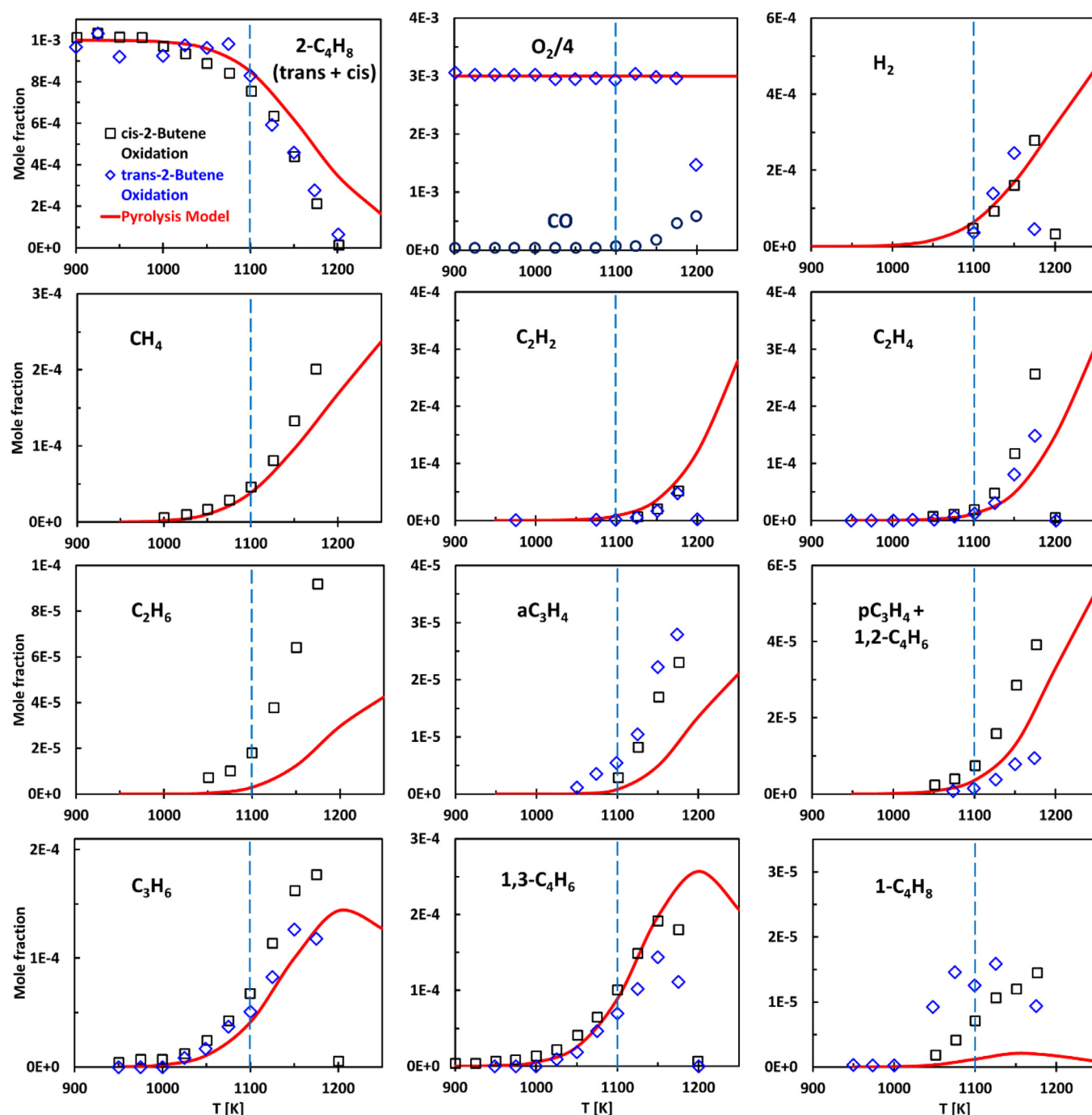


Fig. 22. Comparison of the model predictions (lines) to the literature data (symbols) of cis- and trans-2-butene oxidation in a jet stirred reactor at  $\Phi = 0.5$ ,  $p = 1$  atm,  $\tau = 70$  ms. [14,15]. The vertical line marks the temperature at which CO production becomes apparent, marking the transition from pyrolysis to oxidation kinetics.

radical distribution, thereby changing the distribution of MWG species.

#### Supporting information (SI)

Table S1 provides the measured temperature–distance profiles of the flow reactor. Tables S2 and S3 summarize the experimental conditions for each set of experiments and the results from these experiments are provided in Tables S4–S15. The expanded results from the sensitivity analysis are provided in Tables S16–S19. Figs. S1–S4 are parity plots that compare the model predic-

tions and experimental results for each experimental condition for 1- and 2-butene pyrolysis. Figs. S5 compares the rate constants for selected reactions on the  $\text{C}_4\text{H}_8$  PES that are determined in this work to those in the literature. Fig. S6 shows the product–time profiles that result from numerical integration of the rate constants on the full and simplified  $\text{C}_4\text{H}_8$  PES. More complete versions of the  $\text{C}_6\text{H}_7$ ,  $\text{C}_6\text{H}_9$ , and  $\text{C}_7\text{H}_{11}$  PES are provided in Figs. S7, S8, and S10, respectively. Fig. S9 shows the predicted apparent rate constants for the  $\text{C}_7\text{H}_{13}$  PES at 1 atm. The rates of production for the toluene formation in 1- and 2-butene pyrolysis at 710 °C and 735 °C are shown in Fig. S11. The mechanism and thermodynamic files are provided as text files.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2016.07.037.

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